



Antepartum Fetal Evaluation

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KEY ABBREVIATIONS

American College of Obstetricians and Gynecologists	ACOG	Modified biophysical profile	mBPP
Amniotic fluid index	AFI	Multiples of the median	MoM
Antiphospholipid antibody syndrome	APLAS	National Center for Health Statistics	NCHS
Assisted reproductive technology	ART	National Institute for Child Health and Human Development	NICHD
Biophysical profile	BPP	Nonstress test	NST
Body mass index	BMI	Perinatal mortality rate	PMR
Central nervous system	CNS	Phosphatidylglycerol	PG
Contraction stress test	CST	Pregnancy-associated plasma protein A	PAPP-A
Deepest vertical pocket	DVP	Rapid eye movement	REM
Fetal breathing movement	FBM	Respiratory distress syndrome	RDS
Fetal growth restriction	FGR	Systemic lupus erythematosus	SLE
Fetal movement counting	FMC	Vibroacoustic stimulation	VAS
Human chorionic gonadotropin	hCG	World Health Organization	WHO
Lecithin/sphingomyelin ratio	L/S ratio		

Antepartum fetal evaluation is a complex balancing act. The goal of evidence-based antepartum fetal evaluation is to decrease perinatal mortality and permanent neurologic injury through judicious use of reliable and valid methods of fetal assessment without acting prematurely to modify an otherwise healthy pregnancy or providing a false sense of well-being in cases of impending morbidity. The opportunity for obstetric care providers to participate in this delicate balance has been made possible by continued advances in our ability to assess the physiologic

well-being of the fetus, concurrent with great improvement in neonatal care and survival.¹ However, despite these advances in antepartum fetal surveillance and the widespread use of antepartum testing programs, the ability of these techniques to prevent intrauterine injury or death remains unproved in many cases.² The focus of this chapter is on antepartum evaluation in the United States and similarly technologically advanced and resource-rich countries, noting that the worldwide problem of still-birth is a vast and compelling area of international interest.³

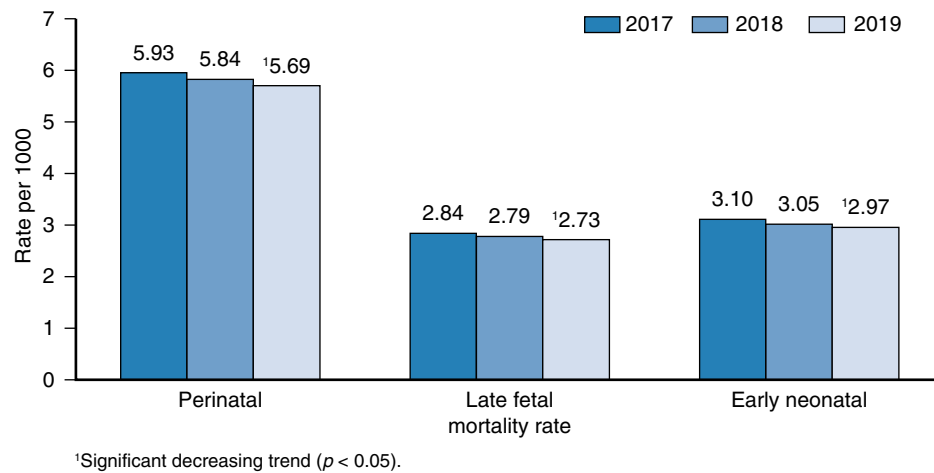


Fig. 27.1 U.S. Trends in Perinatal Mortality Rates Over Time, 2017 to 2019. (Data from the Centers for Disease Control and Prevention/National Center for Health Statistics, National Vital Statistics System. <https://dx.doi.org/10.15620/cdc.112643>; 2022.)

DEFINING THE PROBLEM OF PERINATAL MORTALITY

Identification of fetuses at risk for perinatal mortality has historically been the goal of antepartum fetal assessment. The implementation of this goal is made more complex by the understanding that (1) long-term neurologic disability is an integrally related and often competing entity to perinatal mortality, and (2) identification of fetal death risk may result in iatrogenic prematurity, which itself is a risk for perinatal mortality.

The National Center for Health Statistics (NCHS) provides two different definitions for perinatal mortality, acknowledging that the variation in definitions and reporting rates among states in the United States and throughout different countries worldwide makes comparisons difficult; an agenda to develop a classification consensus has been the focus of a number of international committees, including the National Institute for Child Health and Human Development (NICHD).³ The NCHS National Vital Statistics Report on fetal and perinatal mortality describes two different definitions for *perinatal mortality rate* (PMR). *Definition I* includes the deaths of infants younger than 7 days of age and fetal deaths of 28 weeks' gestation or more per 1000 live births plus fetal deaths, whereas *definition II* is more comprehensive and includes infant deaths of younger than 28 days of age and fetal deaths of 20 weeks or more per the same denominator.^{4,5} The definition of PMR described by the American College of Obstetricians and Gynecologists (ACOG) includes the number of fetuses and live births weighing at least 350 g if gestational age is unknown.⁶ The term *fetal death* is used in these definitions and hereafter in this chapter rather than *stillbirth*, *spontaneous abortion*, or *miscarriage*.

The PMR began to fall in the United States in 1965, with a relative plateau in the mortality rate from 2000 to 2016, and notably began to decline again from 2017 to 2019 (Fig. 27.1).^{4,5,7} Using NCHS definition I, the PMR reported in 2019 was 5.7 per 1000, and fetal deaths accounted for about 50% of all perinatal mortality in the United States.⁴ The PMR varies greatly by maternal race and ethnicity (Fig. 27.2). In 2019, rates (per 1000) were 4.7 for non-Hispanic Whites, 5.1 for Hispanics, and 9.9 for non-Hispanic Black women. Although the PMR declined most for non-Hispanic Black women between 2006 and 2019 (from 11.76 to 9.9), this group continues to experience

double the PMR of non-Hispanic White women. The significantly greater PMR in pregnant individuals at risk for exposure to systemic racism results from higher rates of *both* neonatal and fetal deaths.^{4,7,8}

Characteristics of Fetal Death

Another way to consider the effect of fetal events on PMR is to look at the infant mortality rate (Fig. 27.3).⁸ Although the infant mortality rate includes all deaths of infants younger than 1 year of age, 50% of all infant deaths occur in the first week of life and 50% of these losses result during the first day of life.⁶ In 2019, the leading causes of infant mortality were congenital malformations (21%), disorders related to short gestation and low birthweight (17%), maternal complications (6%), sudden infant death syndrome (6%), and unintentional injuries (6%).⁸ Clearly, **antepartum and perinatal events play an important role in infant mortality.**

Causes of Fetal Death

In addition to declining frequency in PMR over time, the overall pattern of perinatal deaths in the United States has changed considerably during the past 50+ years. Investigators in the 1980s to 1990s^{9,10} who studied longitudinal cohorts from the mid- to late 20th century suggested that etiologies for antepartum fetal deaths could be categorized as chronic asphyxia/placental insufficiency; congenital malformations; superimposed complications of pregnancy (such as Rh isoimmunization, placental abruption, and fetal infection—many as a consequence of preterm premature rupture of membranes); and unexplained cause. The decline in the fetal death rates in these cohorts over time was attributed to the prevention of Rh sensitization, antepartum and intrapartum fetal surveillance, improved detection of fetal growth restriction (FGR) and fetal anomalies with ultrasound, and improved care of maternal diabetes mellitus and preeclampsia. The role of antenatal diagnosis and management of congenital malformations and aneuploidy is obviously critical to the goal of reducing perinatal morbidity and mortality and is discussed separately (see Chapters 10 and 12).

Undiagnosed FGR, in particular, has been identified by many investigators as a potentially preventable contributor to fetal death rates, especially in the gestational period between 28 and 36 weeks.^{10,11} Despite a marked fall in unexplained fetal deaths in the cohorts studied during the second half of the last century, this category was

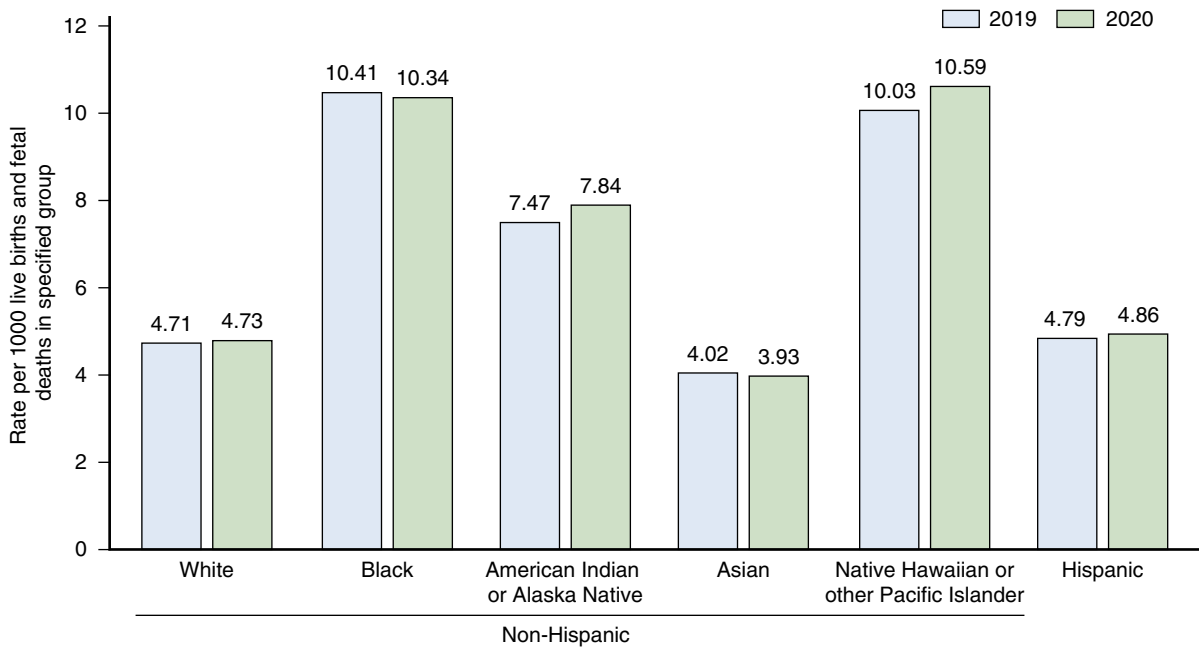


Fig. 27.2 U.S. Fetal Mortality Rates by Maternal Race and Ethnicity, 2020. (Data from the Centers for Disease Control and Prevention/National Center for Health Statistics, National Vital Statistics System, August 2022.)

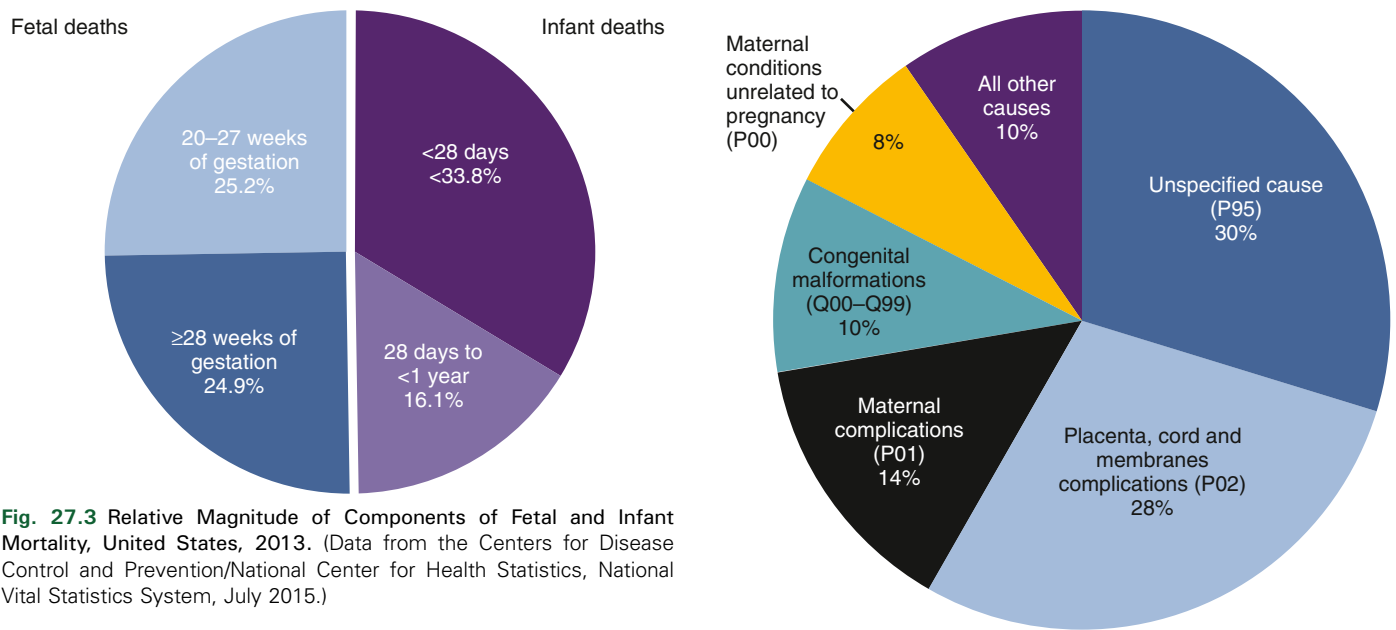


Fig. 27.3 Relative Magnitude of Components of Fetal and Infant Mortality, United States, 2013. (Data from the Centers for Disease Control and Prevention/National Center for Health Statistics, National Vital Statistics System, July 2015.)

used for more than 25% of all stillbirths and is still the categorization used to describe 30% of fetal deaths in the latest US National Vital Statistic Report on the subject.¹² Numerous population-based analyses of the causes of fetal death in the United States and other similarly developed nations have since been performed, including serial evaluations by the Stillbirth Collaborative Research Network, which used rigorous methods to classify as many “unexplained” fetal deaths as possible.¹³ This group and others concluded that **unspecified causes and complications related to the placenta, cord, or membranes are the leading classifications for fetal death used in a modern high resource population at 30% and 28%, respectively, followed by maternal complications of pregnancy and congenital anomalies at**

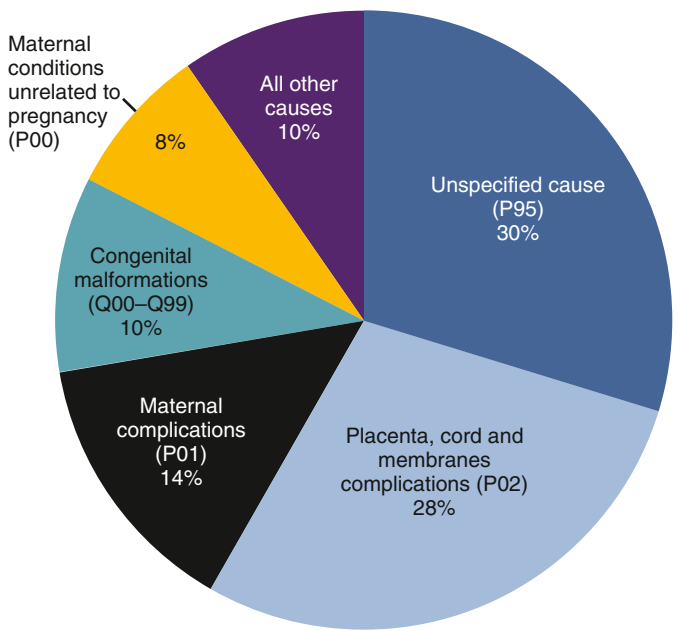


Fig. 27.4 Fetal Deaths by Selected Causes, 2014. Codes in parentheses are cause of death codes as classified by the international statistical classification of diseases and related health problems, tenth revision. (Data from the Centers for Disease Control and Prevention/National Center for Health Statistics, National Vital Statistics System, October 2016.)

14% and 10%, respectively (Fig. 27.4); in addition, fetal deaths late in gestation are more likely than earlier gestation fetal deaths to have no identifiable etiology.¹⁴ Regarding our ability to whittle away at the number of unexplained fetal deaths, and thus possibly increase the proportion of deaths that is potentially preventable, many leaders

in the field have put out a call to action for standardized and rigorous methodology in clinical and research settings when classifying and cataloging each fetal death that occurs in high-resource settings.¹³⁻¹⁵

The ability of our current methods of surveillance to make an impact on perinatal mortality will depend on the ability of available tests to predict and predate injury and on the use of obstetric interventions to prevent adverse outcomes. Since the decline in fetal deaths attributable to factors largely considered identifiable and modifiable during routine modern prenatal care, including congenital malformations and Rh isoimmunization, the concept of further progress toward identifying and reducing other causes of *potentially preventable* fetal deaths has emerged as an important goal. In one British series,¹⁶ obstetric and pediatric assessors reviewed the circumstances surrounding each case of perinatal death to identify any “avoidable” factors that may have contributed to the death. Of the 309 perinatal deaths in this population (half fetal and half in the first week of life), 59% were considered to have had avoidable factors, including 74% of normal-birthweight infants with no fetal abnormalities and no maternal complications. Most avoidable factors were found to be obstetric rather than pediatric, maternal, or social. Failures to respond appropriately to abnormalities during pregnancy and labor—including results from the monitoring of fetal growth or intrapartum fetal well-being, significant maternal weight loss, or reported reductions in fetal movement—constituted the largest groups of avoidable factors. This characterization of avoidable factors that contribute to perinatal death has been confirmed in additional studies, including one by the US Stillbirth Collaborative Research Network, which conducted a comprehensive analysis of 512 fetal deaths and characterized 22% as potentially preventable.¹⁴ This “potentially preventable” classification was made on the basis of features that could theoretically be detected in an otherwise viable (>24 weeks and 500 g, no lethal anomalies) fetus on the basis of maternal and fetal risk factors and testing. Of those fetal deaths categorized as potentially preventable, the majority

shared a common feature of impaired placental function—a condition theoretically amenable to detection via fetal growth surveillance and perhaps antepartum fetal testing.

Timing of Fetal Death

Another way to classify fetal deaths may be to differentiate those that occur during the antepartum period and those that occur during labor, or intrapartum deaths. **Antepartum fetal death is much more common than intrapartum fetal death, and unexplained fetal death occurs far more commonly than unexplained infant death.**^{4,17} In a population-based study in the United States in 2007, the antepartum fetal death rate was 3.7 per 1000, compared with 0.6 per 1000 intrapartum fetal deaths.¹⁷ Although most fetal deaths occur before 32 weeks’ gestation, in planning a strategy for antepartum fetal monitoring, the risk for fetal death must be examined in the population of women who are still pregnant at that point in pregnancy. The understanding that all pregnancies at a given gestational age and at each subsequent point until delivery are at risk for fetal death has led to the development of a metric with which to examine fetal death rates per gestational age, the *prospective fetal mortality rate*.⁴ Rather than compute the fetal mortality rate as the number of fetal deaths per 1000 live births or fetal deaths at a given gestational age, the prospective fetal mortality rate is defined as the number of fetal deaths per 1000 live births or fetal deaths at a given gestational age or greater (Fig. 27.5). When this approach is taken, the data would suggest that fetuses at 40 to 41 weeks are at a threefold greater risk and those at 42 or more weeks are at a 12-fold greater risk for intrauterine death than fetuses at 28 to 31 weeks. The risks are even higher in multiple gestations as pregnancy progresses. For twin gestations, the optimal time for delivery to prevent late-gestation perinatal deaths is before 39 weeks, and for triplets, 36 weeks.^{18,19} The issue of timing is also illustrated by a cohort study of more than 75,000 singleton pregnancies with fetal growth restriction, the focus of which was to find the

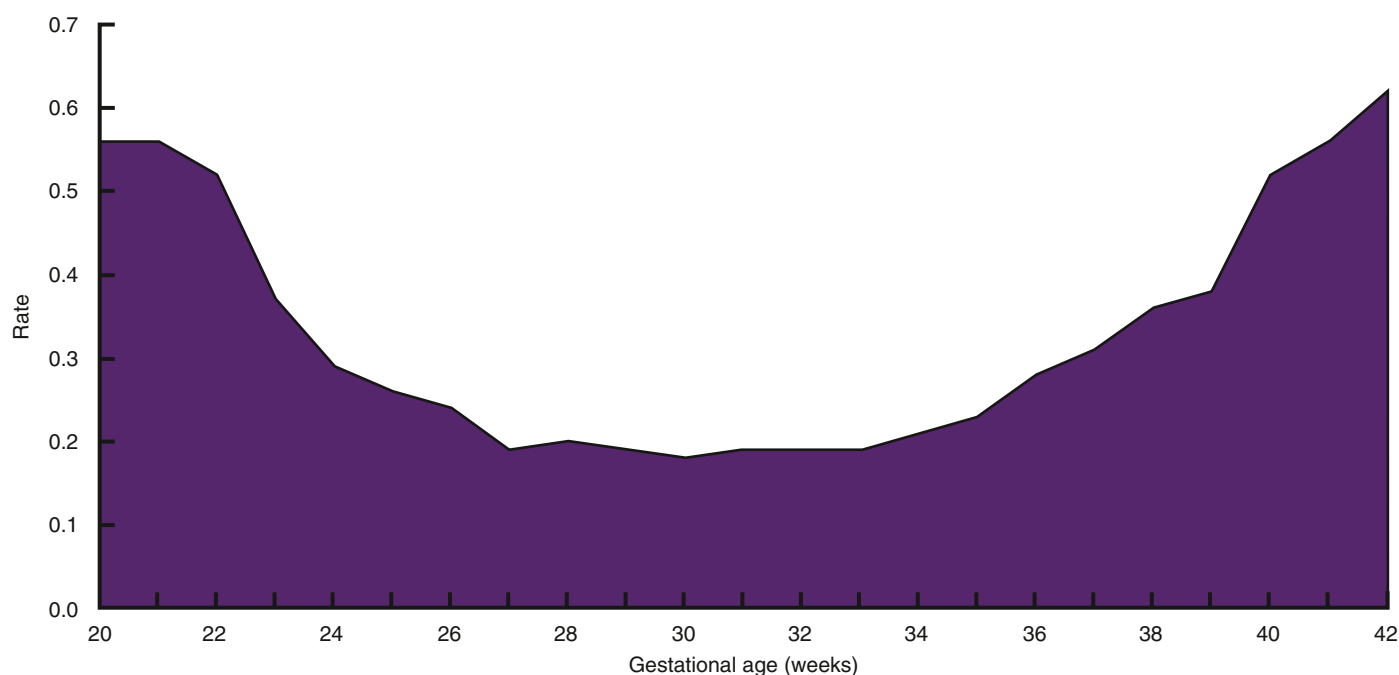


Fig. 27.5 Prospective Fetal Mortality Rate by Week of Gestation, 2013. The prospective fetal mortality rate is the number of fetal deaths at a given gestational age per live births and fetal deaths at that gestational age or greater. (Data from the Centers for Disease Control and Prevention/National Center for Health Statistics, National Vital Statistics System, July 2015.)

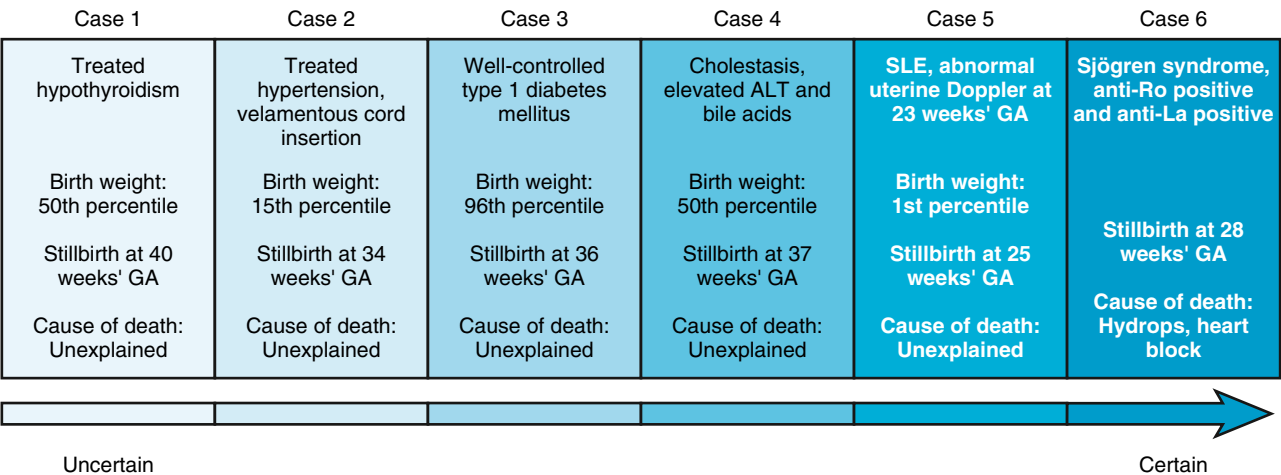


Fig. 27.6 Continuum of Certainty in Pathophysiology of Cause of Fetal Death. Progressing from left to right on the continuum, levels of certainty increase as to the role of the pathophysiology of a particular condition in causing the fetal death. *ALT*, Alanine aminotransferase; *GA*, gestational age; *SLE*, systemic lupus erythematosus. (Courtesy Professor Gordon Smith. Modified from Reddy UM, Goldenberg R, Silver R, et al. Stillbirth classification: developing an international consensus for research. Executive summary of a National Institute of Child Health and Human Development workshop. Stillbirth classification of cause of death. *Obstet Gynecol.* 2009;114:901-914.)

point at which the competing risks of fetal death and neonatal death were in balance, in order to inform delivery decisions. In this cohort, the balance point was 32 to 34 weeks.²⁰

Identifying Those at Risk

Some risk factors have a clear etiologic relationship to fetal compromise and death, such as exposures to teratogens or maternal conditions that alter the fetal environment or blood supply or content. Other risk factors—such as epidemiologic factors that include maternal age, race, and body habitus—have a perhaps more complex and less well-understood link to fetal death risk (Fig. 27.6).³ Common risk factors for fetal death in the United States are listed in Table 27.1.^{2,21,22} Many of these conditions can coexist in individual patients, which makes assessment of the contribution of each factor to perinatal mortality a challenge. A newer concept of looking at individual risk factors as part of a “triple risk model” may prove useful in making sense of this challenge, similar to that used to understand contributors to sudden infant death syndrome (SIDS). In this model, proposed by Warland and Mitchell,²³ an interplay exists among maternal, fetal, and placental factors and a stressor. They posit that whereas these factors in isolation may be insufficient to cause fetal death, they may prove lethal in combination (Fig. 27.7). Also necessary to consider is the contribution of these conditions to fetal injury that results in liveborn children with permanent neurologic compromise; this has yet to be determined but is an important alternative outcome to perinatal mortality that deserves further study.²

SELECT ANTENATAL CONDITIONS THAT MAY CONTRIBUTE TO INCREASED RISK OF FETAL DEATH

Maternal Characteristics

Maternal Age

Multiple investigators²⁴⁻²⁶ have found a progressively increased risk of stillbirth with advancing maternal age with primiparous persons having a higher risk of stillbirth than multiparous persons. In 2020,

TABLE 27.1 Common Risk Factors for Fetal Death in the United States

Risk Factor	Prevalence (%)	Odds Ratio
All pregnancies	—	1.0
Low-risk pregnancies	80	0.86
Obesity (reference: BMI <26):		
BMI 30–34.9	20	1.7
BMI 35–39.9	11	2.0
BMI >40	10	2.5–3.2
Nulliparity compared with second pregnancy	40	1.2–1.6
Fourth child or greater compared with second	11	2.2–2.3
Maternal age (reference: <35 years):		
35–39 years	15–18	1.3–2.2
≥40 years	2	1.8–3.3
Multiple gestation:		
Twins	2.7	1.0–2.2
Triplets or greater	0.14	2.8–3.7
Oligohydramnios	2	4.5
Polyhydramnios	1	1.1-2.9
Assisted reproductive technologies (all)	1–5	1.2–5.0
Abnormal serum markers:		
First-trimester PAPP-A <5%	5	2.2–9.2
Two or more second-trimester markers	0.1–2	4.2–21.9
Intrahepatic cholestasis	<0.1	1.8–5.0
Renal disease	<1	2.2–30
Systemic lupus erythematosus	<1	6–20
Antiphospholipid syndrome	<1	1.2–2.9
Smoking	10–20	1.7–3.0
Alcohol use (any)	6–10	1.2–2.7

Continued

TABLE 27.1 Common Risk Factors for Fetal Death in the United States—cont'd

Risk Factor	Prevalence (%)	Odds Ratio
Illicit drug use	2–4	1.2–3.0
Low education and socioeconomic status	30	2.0–7.0
Fewer than four antenatal visits ^a	6	2.7
Black race (via exposure to racism; reference: White)	15	2.0–2.2
Hypertensive disorders	6–10	1.5–4.4
Diabetes	2–5	1.5–7.0
Large for gestational age (>97% without diabetes)	12	2.4
Fetal growth restriction (%):		
<3	3.0	4.8
3–10	7.5	2.8
Previous growth-restricted infant	6.7	2.0–4.6
Previous preterm preeclampsia	1	2.3–5.6
Previous preterm birth with growth restriction	2	4.0–8.0
Decreased fetal movement	4–8	4.0–12.0
Previous stillbirth	0.5	2.0–10.0
Previous cesarean section	22–25	1.0–1.5
Postterm pregnancy compared with 38–40 weeks		2.0–3.0
41 weeks	9	1.5
42 weeks	5	2.0–3.0

^aFor stillbirths, 37 weeks' gestation.

BMI, Body mass index; PAPP-A, pregnancy-associated plasma protein A.

Modified from Signore C, Freeman RK, Spong CY. Antenatal testing: a reevaluation. Executive Summary of a Eunice Kennedy Shriver National Institute of Child Health and Human Development Workshop. *Obstet Gynecol.* 2009;113:687-701; Fretts RC. Stillbirth epidemiology, risk factors, and opportunities for stillbirth prevention. *Clin Obstet Gynecol.* 2010;53:588-596; and Indications for outpatient antenatal fetal surveillance. ACOG Committee Opinion No. 828. American College of Obstetricians and Gynecologists. *Obstet Gynecol.* 2021;137:e177-97.

the stillbirth rates were highest for women younger than 15 years old and 45 years and older and lowest for women aged 25 to 34 years.⁴ Women 35 to 39 years old and without medical conditions have a 1.28-fold increased risk ratio (RR) (95% confidence interval [CI], 1.17 to 1.41) of stillbirth at term (37 to 41 weeks) compared with women younger than 35 years. Women without medical conditions who are age 40 and older have a 1.79-fold increased RR (95% CI, 1.52 to 2.10) of stillbirth at term (37 to 41 weeks) compared with women younger than 35 years.²⁴ This highlights the point that otherwise healthy older gravidas face an increase in fetal death risk compared with younger women.

Maternal Race and Ethnicity

Race is a social rather than a biological construct, and the effects of racism (structural, institutionalized, and interpersonal) and biases (implicit and explicit) are implicated in many health inequities; these

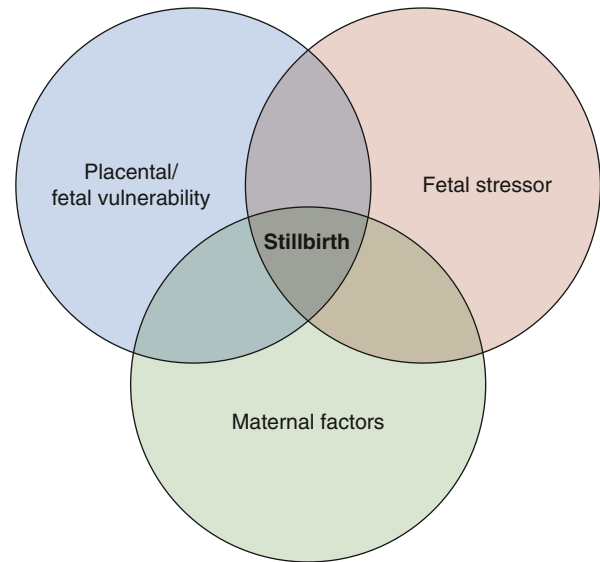


Fig. 27.7 Triple Risk Model. An interplay of maternal factors, fetal/placental factors, and a stressor contribute to fetal death. These factors in isolation may be insufficient to cause fetal death but may prove lethal in combination. (From Warland J, Mitchell EA. A triple risk model for unexplained late stillbirth. *BMC Pregnancy Childbirth.* 2014;14:142.)

are more likely than race to be related to elevated risk.²⁷ Risks of fetal death and causes of fetal death vary by exposure to racism and bias and the downstream effects of that exposure on women's baseline health, pregnancy health, and receipt of care.²² Factors associated with increased rates of fetal death among pregnant women exposed to systemic racism compared with White women include disparities in socioeconomic status, access to health care, and preexisting medical conditions, though increased stillbirth rates in some studies have been shown to be independent of comorbid conditions.^{25,28}

In 2020, the rates of fetal mortality per 1000 live births and fetal deaths (different from the perinatal mortality rates reported earlier, in that fetal mortality includes fetal deaths between 20 weeks' gestation and birth) were 3.93 in non-Hispanic Asian women, 4.73 in non-Hispanic White women, 4.79 in Hispanic women, 7.84 in non-Hispanic single-race American Indian or Alaska Native women, 10.34 in non-Hispanic Black women, and 10.59 in non-Hispanic single-race Native Hawaiian or other Pacific Islander women.⁴ The greatest disparities in fetal death rates by race are earlier in the preterm period compared with later in gestation. At 20 to 27 weeks, the 2020 fetal death rate per 1000 was 2.32 for non-Hispanic White women compared with 5.72 for Black women, whereas the rates at 28 weeks or greater were 2.41 and 4.68, respectively.⁴ The study of racial and ethnic disparities in fetal death rates worldwide, as well as how these intersect with sociodemographic features and what remediating steps could be taken, has been identified as a critically overdue initiative (Fig. 27.8).

Socioeconomic Factors, Prenatal Care, and Substance Abuse

Poor access to prenatal care and poor underlying health and nutrition have been linked to increased risk for fetal death in both developing and developed nations. As with other sociodemographic risk factors, these potential influences on fetal death risk are difficult to quantify and may be additive to other high-risk conditions.²⁹ Smoking and abuse of alcohol or illicit drugs represent potentially modifiable risk factors for fetal death. Although these behaviors are attractive

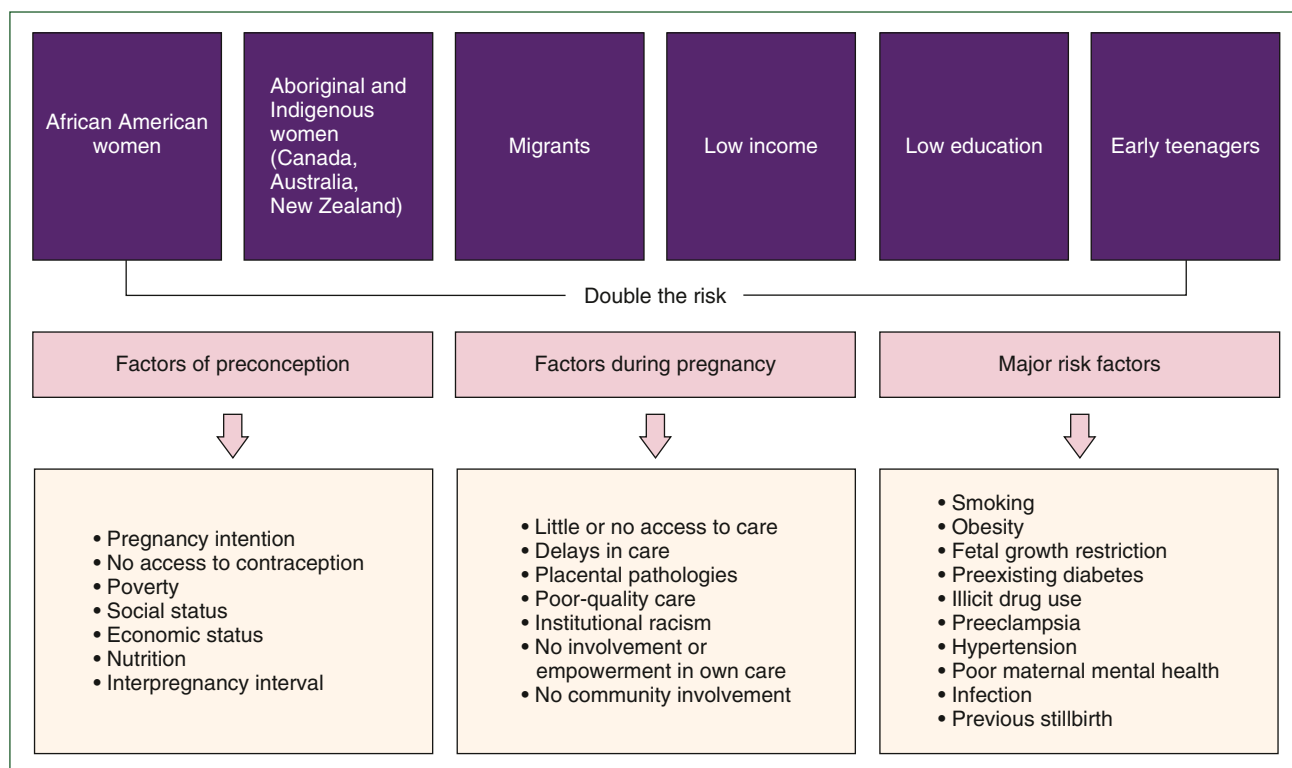


Fig. 27.8 Disparity in Prevalence of Fetal Death Based on Sociodemographic Factors. Complex relationships and associations inform disparity in fetal death rates across populations exposed to systemic racism and other negatively impactful social determinants of health. (From Flenady V, Wojcieszek AM, Middleton P, et al. Lancet Ending Preventable Stillbirths study group. Stillbirths: recall to action in high-income countries. *Lancet*. 2016;387:691-702.)

candidates for fetal death prevention through counseling and modification of intake, prospective trials of behavior modification strategies have generally been underpowered for detection of a difference in fetal death with these interventions.²²

Maternal Comorbidities

Obesity

Prepregnancy obesity is associated with increased perinatal mortality, especially in late gestation. This has been demonstrated in several large series, including a meta-analysis of 38 studies that included more than 3 million women,³⁰ and the risk of stillbirth rises with increasing obesity. Compared with women with a body mass index (BMI) of less than 30, after controlling for characteristics including maternal age, nulliparity, and other comorbid conditions, the hazard ratio for stillbirth is 1.71 for prepregnancy BMI 30.0 to 34.9 kg/m²; 2.00 for BMI 35.0 to 39.9 kg/m²; 2.48 for BMI greater than 40; and 3.16 for women with BMI equal to or greater than 50 kg/m².³¹ The stillbirth risk associated with obesity appears to increase with gestational age. For example, women with a BMI equal to or greater than 30 kg/m² have an adjusted hazard ratio of 3.5 (95% CI, 1.9 to 6.4) at 37 to 40 weeks and 4.6 (95% CI, 1.6 to 13.4) at greater than 40 weeks compared with women of normal weight (BMI 18.5 to 25 kg/m²).³²

Diabetes Mellitus

Although historically, insulin-dependent diabetes has been a major risk factor for fetal death, the fetal death rate in women with optimal

glycemic control now approaches that of those without diabetes.^{2,33}

Stillbirth occurs most commonly in women with poor glycemic control and in those with polyhydramnios, fetal macrosomia, or declining insulin requirements.³⁴ The overall stillbirth rate in a cohort of women with pregestational diabetes was 13.9 per 1000 pregnancies (95% CI, 9.7 to 19.9), and women with pregestational diabetes had a significantly higher stillbirth rate at all gestations after 32 weeks.³⁵ In patients with diabetes who have renal disease, vascular disease, fetal growth restriction, or concomitant hypertension, stillbirth may occur as early as the late second trimester.³⁶ No evidence suggests that gestational diabetes controlled by diet alone is associated with increased rates of intrauterine fetal death, while data are mixed regarding any increase in the odds of fetal death attributable to gestational diabetes requiring treatment with hypoglycemic agents.²

Hypertensive Disorders

Studies have shown conflicting evidence regarding whether fetal death rates with well-controlled preexisting hypertension are comparable with those in the general population or increased. When chronic hypertension is complicated by superimposed preeclampsia, however, the risk is increased more than fourfold (RR, 4.4; 95% CI, 2.2 to 8.8).³⁷ The estimated rates of stillbirth in pregnancies complicated by gestational hypertension with and without severe range blood pressure are 12 to 29 per 1000 (OR, 1.8 to 4.4) and 9 to 51 per 1000 (OR, 1.2 to 4.0), respectively.^{5,37} The increased risk for perinatal mortality associated with hypertension is most often related to sequelae of placental

insufficiency, which include FGR and oligohydramnios. Gestational hypertension, especially preeclampsia with severe features or eclampsia, may be associated with fetal death through placental and coagulation-related pathways, including placental abruption.³

Thrombophilia

In general, no demonstrable link has been found between inherited thrombophilia and risk for fetal death.^{3,21} The presence of circulating maternal antiphospholipid antibodies—in particular lupus anticoagulant, anticardiolipin antibodies, and anti- β_2 -glycoprotein I antibodies—in the antiphospholipid antibody syndrome (APLAS) has been associated with a variety of adverse pregnancy outcomes, including fetal loss.³⁸ The risk of stillbirth in women with antiphospholipid syndrome diagnosed by established criteria³⁸ is related to the number of positive antibodies, with rates of 217 per 1000 pregnancies with a single antibody positive result compared with 364 per 1000 pregnancies with multiple antibody positive results (aOR 2.67; 95% CI, 1.22 to 2.94). Such risk is independent of treatment with low-dose aspirin and low-molecular-weight heparin.³⁹ The mechanism of these adverse outcomes remains unclear but likely includes inflammation, thrombosis, and placental infarction.³

Intrahepatic Cholestasis

The mechanism of fetal death in individuals with intrahepatic cholestasis of pregnancy has been hypothesized in some cases to be related to sudden fetal cardiac arrhythmia; however, **the cause of fetal death remains unknown, and timing and disease characteristics associated with impending fetal death remain unpredictable.** Fetal deaths in these pregnancies are not preceded by signs of placental insufficiency, such as growth restriction, and normal fetal heart rate tracings proximal to fetal death (i.e., within 24 hours) have often been reported. There appears to be a positive correlation between maternal levels of bile acids and the risk of stillbirth, particularly with total bile acid levels greater than 40 mmol/L⁴⁰ or, in other studies, greater than 100 mmol/L.^{41,42} For singleton pregnancies with intrahepatic cholestasis, the prevalence of stillbirth was 1.3 per 1000 births with total bile acids of less than 40 micromole/L; 2.8 per 1000 births with total bile acids of 40 to 99 micromole/L; and 34.4 per 1000 births with total bile acids of 100 micromole/L or greater.⁴²

Renal Disease and Systemic Lupus Erythematosus

With chronic maternal renal disease, the degree of impairment of renal function appears to be the major determinant of pregnancy outcome, with a particularly high risk seen among women who require dialysis for end-stage renal disease.^{43,44} Risk of fetal death among women with chronic renal disease has been estimated to occur in 15 to 200 per 1000 pregnancies.^{2,45} Mild, moderate, and severe renal insufficiency can be defined as serum creatinine 0.9 to 1.4 mg/dL, greater than 1.4 but less than or equal to 2.5 mg/dL, and greater than 2.5 mg/dL, respectively.⁴³ Women with mild renal disease have a lower risk of adverse pregnancy outcomes and do not appear to be at increased risk for stillbirth.

As with maternal renal disease, the prognosis for fetal outcome in women with systemic lupus erythematosus (SLE) is dependent on disease state and comorbid conditions, including hypertension, antiphospholipid autoantibodies, and renal involvement.⁴⁶ The risk of stillbirth among women with lupus is estimated to be 40 to 150 per 1000 pregnancies² with the higher end of the range likely contributed to by coexisting hypertensive disorders or the presence of

antiphospholipid antibodies.^{46,47} The prognosis for fetal survival in pregnancies complicated by both maternal renal disease and maternal SLE has improved over time with advances in therapies that promote disease quiescence for both of these conditions.

Sickle Cell Disease

Controversy exists as to whether pregnancy complicated by sickle cell disease is at increased risk for fetal demise. A systematic review and meta-analysis of 19 studies from nine different countries found an increased risk of stillbirth associated with maternal sickle cell disease (81 per 1000; pooled OR, 4.05; 95% CI, 2.69 to 6.32; $P < .001$), which was observed in both low-income (OR, 3.59; 95% CI, 2.59 to 6.32; $P < .001$) and high-income countries (OR, 5.09; 95% CI, 2.38 to 10.90; $P < .001$).⁴⁸ Conversely, a retrospective statewide cohort study of a contemporary North American cohort⁴⁹ found no increased rate of stillbirth in pregnancies complicated by sickle cell disease. The authors hypothesized that the low rates of stillbirth may reflect improved antenatal surveillance and management compared with previous studies.

Antiphospholipid Syndrome

The risk of stillbirth in pregnancies complicated by antiphospholipid syndrome^{38,39} is related to the number of positive antibodies, with rates of 217 per 1000 pregnancies with a single antibody positive result compared with 364 per 1000 pregnancies with multiple antibody-positive results (aOR 2.67; 95% CI, 1.22 to 2.94).³⁹ This risk is independent of treatment with low-dose aspirin and low-molecular-weight heparin.³⁹

Obstetric Factors

Obstetric History and Assisted Reproductive Technology

Multiple aspects of a pregnant individual's obstetric and fertility history may be associated with risk for fetal death in a current pregnancy, including parity, use of assisted reproductive technology (ART),⁵⁰ and history of prior adverse obstetric outcomes. Both nulliparity and high parity are associated with an increased risk for fetal death compared with low multiparity (one, two, or three prior births).⁵¹ This association is likely mediated through a variety of sociodemographic risk factors related to overall health and interconception health status, although studies have confirmed the association between parity and fetal death after controlling for several social and medical comorbidities. A history of prior adverse pregnancy outcomes—including fetal growth restriction, preterm birth, and fetal death—marks a current pregnancy as at risk for fetal death.

A history of a neonate who was small for gestational age (defined as birth weight 2 standard deviations or more below the mean for gestational age) without associated anomalies in the first pregnancy is associated with an increased risk of stillbirth in the subsequent pregnancy⁵² after adjusting for covariables associated with the risk of stillbirth. The increased risk of stillbirth is primarily in the preterm period and persists even if the second pregnancy is the appropriate size for gestational age.⁵² The risk is inversely related to the gestational age at the first small-for-gestational-age birth: aOR was 2.1 (95% CI, 1.6 to 2.8) for term deliveries; 3.4 (95% CI, 2.1 to 5.6) for gestational ages between 32 and 36 weeks; and 5.0 (95% CI, 2.5 to 9.8) for gestational ages younger than 32 weeks. The corresponding rates of stillbirth were 4.8 per 1000, 9.5 per 1000, and 19 per 1000, respectively.⁵² Similar findings were reported by other investigators, with severity of growth restriction in the antecedent pregnancy related to risk of stillbirth in the second pregnancy.⁵³

A history of preeclampsia is associated with an increased risk of stillbirth in the next pregnancy, even without recurrent preeclampsia,^{53,54} and the risk of stillbirth is inversely related to the gestational age at delivery in the previous pregnancy with the adverse obstetric outcome. In a large nationwide study,⁵³ the odds ratio for stillbirth in a second pregnancy was 2.3 (95% CI, 1.5 to 3.6; rate: 11 per 1000) if the first pregnancy was complicated by preeclampsia with delivery at 32 to 36 weeks; 3.8 (95% CI, 1.8 to 7.8; rate: 18 per 1000) if the preeclampsia occurred with delivery at 28 to 32 weeks; and 5.6 (95% CI, 1.3 to 23.2; rate: 23 per 1000) if preeclampsia occurred with delivery at 20 to 27 weeks.

Compared with women without a history of stillbirth, those with a previous stillborn infant are 4.83 times (95% CI, 3.77 to 6.18) more likely to have a subsequent stillbirth.⁵⁵ The risk of recurrent stillbirth may be increased as high as 10-fold depending on maternal race and characteristics of the previous stillbirth, such as etiology, gestational age, and presence of fetal growth restriction.⁵⁶ The etiology of a previous stillbirth affects the ability of antenatal fetal surveillance to prevent recurrences. However, for many cases of stillbirth, the etiology is unknown.³

Multiple Gestations. The higher rate of perinatal mortality in multiple gestations compared with singletons is related to both complications unique to multiple gestations, such as twin-to-twin transfusion syndrome, and more general complications, such as fetal abnormalities and growth restriction.²² Chorionicity is of paramount importance in determining fetal risk, and rates of adverse outcomes are higher among monochorionic twins.² A systematic review¹⁹ found that the risk of stillbirth increased in all twins with advancing gestational age, but it was significantly greater in monochorionic than dichorionic twins. In dichorionic twins, the stillbirth rates were 0.8 per 1000 at 35^{0/7} to 35^{6/7} weeks, 1.5 per 1000 at 36^{0/7} to 36^{6/7} weeks, 3.4 per 1000 at 37^{0/7} to 37^{6/7} weeks, and 10.6 per 1000 at 38^{0/7} to 38^{6/7} weeks. In monochorionic diamniotic twins the stillbirth rates were 0.9 per 1000 at 34^{0/7} to 34^{6/7} weeks, 2.8 per 1000 at 35^{0/7} to 35^{6/7} weeks, 4.5 per 1000 at 36^{0/7} to 36^{6/7} weeks, and 9.6 per 1000 at 37^{0/7} to 37^{6/7} weeks. Additionally, many women who carry more than one fetus have maternal risk factors for increased perinatal mortality, including advanced maternal age and use of ART, and are subject to the development of complications such as preeclampsia and preterm delivery.² Optimal timing of delivery between 37 and 38 weeks has been considered for twins, compared with 39 to 40 weeks among singletons, because of the increased rate of late fetal death in this group.^{21,57}

Early Pregnancy Markers

First- and second-trimester serum markers for aneuploidy, when abnormally low or elevated, have been associated to varying degrees with adverse perinatal outcomes even in the absence of aneuploidy. Biophysical uterine factors have also been studied in this light. Markers found to be associated with increased risk of stillbirth include reduced first-trimester levels of pregnancy-associated plasma protein A (PAPP-A) and elevated second-trimester levels of maternal serum α -fetoprotein (MSAFP) and inhibin A.

Pregnancies complicated by first trimester pregnancy-associated plasma protein A (PAPP-A) levels less than or equal to the fifth percentile (0.415 multiples of the median [MoM]) have a 2.15-fold to 9.2-fold increased risk of stillbirth at greater than 24 weeks, with rates of 5.8 to 20 per 1000, respectively.⁵⁸ The lower the PAPP-A level, the higher the risk of stillbirth that has been associated with it.⁵⁸

Unexplained elevated second-trimester MSAFP is associated with increased risk of stillbirth with RR estimates ranging from 3.4 to 21.9 per 1000.⁵⁸ Similar to PAPP-A, the risk of stillbirth increases with increasing MSAFP level. The increased risk of stillbirth after 24 weeks persisted when preeclamptic and low-birthweight participants were excluded. The association with stillbirth strengthens if elevated inhibin A equal or greater than 2.0 MoM is associated with other abnormal markers.⁵⁸

The sensitivity and positive predictive values of these markers for fetal death are still under investigation. The pathophysiologic link between these markers and adverse outcomes is an unclear but likely variable and most plausibly involves abnormal placental attachment or function.⁵⁹

Amniotic Fluid Abnormalities

The predictive value of either oligohydramnios or polyhydramnios for adverse pregnancy outcomes, in particular fetal death, typically lies in their association with other abnormal conditions, such as maternal diabetes mellitus, hypertensive disorders, rupture of membranes, fetal growth restriction, or fetal anomalies. The stillbirth rate in pregnancies complicated by oligohydramnios, defined as a single deepest vertical pocket less than 2 cm, is estimated at 14 per 1000.² Isolated oligohydramnios is more frequent at term and is commonly considered evidence of placental insufficiency.¹⁴

The increased risk of stillbirth associated with polyhydramnios has mainly been attributed to higher incidence of fetal anomalies. However, a retrospective cohort study of nonanomalous births complicated by polyhydramnios⁶⁰ found that while the risk of stillbirth is 1.14 per 1000 at 32 weeks, an increase to 2.91 per 1000 at 39 weeks (after excluding pregnancies with pregestational or gestational diabetes mellitus) suggests some independent association between apparently “idiopathic” polyhydramnios and increased stillbirth risk. Evaluation of amniotic fluid volume (AFV) as a marker of long-term fetal health status is a mainstay of antepartum fetal evaluation.²

Fetal Growth Restriction

Fetal growth restriction is a well-known risk factor for perinatal death that has historically been underrecognized before fetal death. The risk of stillbirth in pregnancies complicated by fetal growth restriction increases with gestational age and is inversely proportional to the percentile of birthweight for gestational age. The stillbirth risk is 2.63 per 1000 for fetuses with an estimated weight that is lower than the 10th percentile, 4.39 per 1000 for an estimated fetal weight that is lower than the fifth percentile, and as high as 5.8 stillbirths per 1000 at-risk fetuses with an estimated weight lower than the 3rd percentile compared with 0.51 per 1000 for fetuses with normal growth. Placental dysfunction is commonly implicated in nonmalformed and chromosomally normal IUGR fetuses.⁶⁰ This topic is reviewed in further detail in Chapter 30.

Late-Term and Postterm Pregnancy

The definition of postterm pregnancy has been reevaluated in the past 2 decades on the basis of the reappraisal of the peak time of fetal risk (see Chapter 29) and the results of the ARRIVE trial,⁶¹ showing favorable outcomes in a low-risk population delivered without other medical indication at 39 weeks. The risk of stillbirth at term increases with gestational age from 0.11 per 1000 ongoing pregnancies at 37 weeks (95% CI, 0.07 to 0.15) to 1.78 per 1000 ongoing pregnancies at 41 weeks (95% CI, 1.52 to 2.07) and to 3.18 per 1000 ongoing

pregnancies at 42 weeks (95% CI, 1.84 to 4.35).⁶² The pathophysiology of increased fetal death risk in the postterm pregnancy is thought to be mediated by impaired placental oxygen exchange and is often associated with oligohydramnios.

Fetal Malformations and Aneuploidies

Pregnancies complicated by major fetal anomalies or aneuploidy, including trisomy 21, are at increased risk of stillbirth, independent of coexisting fetal growth restriction. The association between anomalies and fetal death was demonstrated in a large retrospective cohort study in which the overall stillbirth rate among fetuses with a major anomaly was 55 per 1000 compared with 4 per 1000 in nonanomalous fetuses, with an adjusted odds ratio of 15.⁶³ The rate of stillbirth was highest in fetuses with congenital cardiac defects. A greater than twofold increased risk of stillbirth has been reported for cardiac anomalies, even in the absence of associated chromosomal anomalies or other major structural anomalies.⁶⁴ The excess stillbirth risk is noted even for anomalies not affecting major vital organs, such as cleft lip and palate.⁶⁵

The stillbirth rates in pregnancies affected by fetal trisomy 21 are estimated at 5% to 10%.⁶⁶ This excess risk is likely related to associated structural abnormalities, fetal growth restriction, or placental dysfunction related to the coexisting placental aneuploidy.

Decreased Fetal Movement

Maternal perception of fetal movements (“kick counts”) is the oldest and most commonly used method to assess fetal well-being. It is hypothesized that decreased fetal movements may be an adaptive response to uteroplacental insufficiency that results in either acute or chronic fetal hypoxemia. Because neither the optimal number of movements nor the duration of monitoring has been determined, studies that assessed the relationship between fetal movement and perinatal mortality have used varying definitions of decreased fetal movement. Nevertheless, decreased fetal movements have been associated with an increased risk of stillbirth⁶⁷ (odds ratio [OR], 2.9 to 4.51) with the rate of stillbirth after reduced fetal movement estimated to be 13 per 1000 episodes.²

Chronic Placental Abruption

Placental abruption has a high-positive association with stillbirth, with estimated risks as high as 4% to 7% in one population-based study.¹⁷ When placental abruption occurs at or near term, delivery is most often recommended. However, in the preterm period with reassuring maternal and fetal status, close observation is reasonable.

Vasa Previa, Velamentous Cord Insertion, and Single Umbilical Artery

Abnormalities of placental shape, single umbilical artery, and abnormalities of umbilical cord insertion (e.g., velamentous cord insertion) are thought to reflect abnormal placental implantation. It is estimated that up to 57% of stillbirths are due to placental abnormalities or insufficiency.¹⁴ Rupture of submembranous fetal vessels that course across the cervical os may result in stillbirth in women with vasa previa.³ In the population-based case-control study of stillbirth conducted by the National Institute of Child Health and Human Development Stillbirth Collaborative Research Network, the odds ratios for stillbirth were 4.80 (95% CI, 2.67 to 8.62) for single umbilical artery and 4.50 (95% CI, 2.18 to 9.27) for velamentous cord insertion.⁶⁸

Conclusions

The best use of available antenatal testing modalities may vary according to the risk profile of each individual pregnancy. Discussion of

condition-specific testing will be undertaken after review of the individual testing modalities described later.

POTENTIAL UTILITY OF ANTEPARTUM FETAL TESTING

Can antepartum fetal deaths and injury be prevented? Before using antepartum fetal testing, the obstetrician must ask several important questions:

1. Does the test provide information not already known by the patient's clinical status?
2. Can the information be helpful in managing the patient?
3. If an abnormality is detected, is a treatment available for the problem?
4. Could an abnormal test result lead to increased risk for the pregnant individual or fetus?
5. Will the test ultimately decrease perinatal morbidity and mortality?

A large body of clinical and research experience suggests that antepartum fetal assessment can have a significant impact on the frequency and causes of antenatal fetal deaths.¹ However, according to several reviews of the benefits and costs of antenatal testing, “strong evidence for the efficacy of antepartum testing is lacking.”⁶⁹ **Unfortunately, few of the antepartum tests commonly used in clinical practice today have been subjected to large-scale prospective and randomized evaluations that can speak to the true efficacy of testing.**²¹ In most cases, the test has been applied and good perinatal outcomes have been observed; therefore the test has gained further acceptance and has been used more widely. In such cases, it is uncertain whether the information provided by the test has accurately led to the improved outcomes or the total program of care has made the difference. When prospective randomized investigations are conducted, large numbers of patients must be studied because many adverse outcomes, such as intrauterine death, are uncommon even in high-risk populations. For example, although several controlled trials have failed to demonstrate improved outcomes with nonstress testing, the study populations ranged from only 300 to 530 subjects.

To determine the clinical application of antepartum diagnostic testing, the predictive value of the tests must be considered. Antepartum fetal tests may be used to screen a large obstetric population to detect fetal disease. In this setting, a test of high sensitivity (test is positive when disease is actually present) is preferable to minimize the risk of missing a patient whose fetus might be compromised. It would be prudent to be willing to overdiagnose the problem—that is, to accept some false-positive diagnoses. In further evaluating the patient whose fetus may be at risk, and when attempting to confirm the presence of disease, a test of high specificity (test is negative when disease is truly absent) is preferable. It is best not to intervene unnecessarily and deliver a fetus who was doing well. In this setting, multiple tests may be helpful. When multiple test results are normal, they tend to exclude disease; however, when all are abnormal, they tend to support the diagnosis of fetal disease.

The prevalence of the abnormal condition has great impact on the predictive value of antenatal fetal tests and the number needed to evaluate with testing and to treat with interventions (delivery) to presumably prevent fetal death. The positive predictive value of an antenatal test of fetal well-being is the likelihood that a patient with an abnormal test truly has an increased risk for fetal death—which is not dependent on the test characteristics themselves but rather on the prevalence of the condition in the population to which the test is applied. The impact of these parameters on the utility of testing was illustrated in a decision analysis of the risks and benefits of antepartum

testing late in pregnancy for women 35 years or older using the McGill Obstetric/Neonatal Database to obtain risk estimates.⁷⁰ In this model, as in practice, the relative benefit of antenatal testing lies in the balance of the number of fetal deaths prevented with the number and type of interventions required to prevent them. In the McGill cohort, a change in fetal death prevalence from 1.5 per 1000 to 5 per 1000 tripled the number of antenatal tests, inductions of labor, and cesarean deliveries required to prevent one additional fetal death. Thus **the number needed to evaluate and treat to prevent one fetal death decreases as risk for fetal death increases in the population being tested.**

In interpreting the results of studies of antepartum testing, obstetricians must consider the application of those study results to their own populations. If the study has been done in a population of patients at great risk, it is more likely that an abnormal test will be associated with an abnormal fetus. If the obstetrician is practicing in a community with patients who are, in general, at low risk, an abnormal test result will more likely be associated with a false-positive diagnosis.

WHAT DO THESE TESTS TELL US ABOUT THE FETUS?

Fetal State

To be able to diagnose suspected fetal compromise using tests of fetal biophysical state, blood flow, and heart rate, we must be able to appreciate how these parameters appear under normal conditions and in response to suboptimal conditions.

Regarding fetal biophysical characteristics, it must be appreciated that during the third trimester, the normal fetus can exhibit marked changes in its neurologic state.⁷¹ Four fetal states have been identified. The near-term fetus spends about 25% of its time in a *quiet sleep state* (state 1F) and 60% to 70% in an *active sleep state* (state 2F). Active sleep is associated with rapid eye movement (REM). The fetus exhibits regular breathing movements and intermittent abrupt movements of its head, limbs, and trunk. The fetal heart rate in active sleep (state 2F) exhibits increased variability and frequent accelerations with movement. During quiet, or non-REM, sleep, the fetal heart rate slows, and heart rate variability is reduced. The fetus may make infrequent breathing and startled movements. Near term, periods of quiet sleep may last 20 minutes, and those of active sleep, about 40 minutes.⁷¹ The mechanisms that control these periods of rest and activity in the fetus are not well established. External factors such as the pregnant individual's activity, her ingestion of drugs, and her nutrition may play a role. Specific factors that may decrease fetal movement in the third trimester include fetal anomalies, particularly central nervous system (CNS) anomalies; maternal exposures, including corticosteroids, sedatives, smoking, and anxiety; low AFV; and decreased placental blood flow due to placental insufficiency.⁷²

When evaluating fetal conditions using the nonstress test (NST) or the biophysical profile (BPP), the clinician must consider whether a fetus not making breathing movements or showing accelerations of its baseline heart rate is in a quiet sleep state or is neurologically compromised. In such circumstances, prolonging the period of evaluation usually allows a change in fetal state, and more normal parameters of fetal well-being will appear.

Regarding regulation of fetal heart rate and blood flow, **fetal adaptation to hypoxemia is mediated through changes in heart rate and redistribution of cardiac output.** However, changes in fetal cardiac output are generally observed during hypoxemia only with coexisting acidemia. In response to sudden hypoxemia, fetal heart rate slowing and increased variability can be observed initially through

vagally mediated chemoreceptor responses. With prolonged hypoxemia (30 to 60 minutes), increasing levels of circulating adrenergic agonists and modulation of vagal activity by endogenous opiates lead to a fetal heart rate return to or rise from the previous baseline.⁷³ Development of acidemia on top of hypoxemia can accelerate the rate of fetal deterioration and amplify the hypoxemia by a shift of the oxyhemoglobin dissociation curve to the right, which further reduces the oxygen-carrying capacity of fetal blood and eventually leads to a redistribution of cardiac output that can be appreciated as a "brain-sparing" effect in the evaluation of fetal blood flow. Redistribution of blood flow in the compromised fetus preferentially preserves perfusion not only to the brain but also to the heart and adrenal glands.⁷³

Fetal movement is a more indirect indicator of fetal oxygen status and CNS function, and decreased fetal movement (DFM) is noted in response to hypoxemia.² However, gestational development of fetal movement must be considered when evaluating fetal well-being as marked by fetal activity. Periods of absent fetal movement become more prolonged as gestation advances, and normal fetuses progressively exhibit longer periods of quiescence as the late second and third trimesters advance. Up to and perhaps longer than 40 minutes of fetal inactivity at 40 weeks may be a normal finding, compared with less than 10 minutes at 20 weeks and less than 20 minutes at 32 weeks.⁷³ Keeping these trends in mind, an abnormal degree or absence of fetal movement can be an appropriate marker for fetal hypoxemia. However, fetal activity levels have been seen in animal studies to adapt to inducement of hypoxemia with a resumption of fetal breathing and body movements after a prolonged period of hypoxemia, especially if induced gradually. Therefore observation of these fetal states during antenatal testing does not guarantee a normoxic fetus.

BIOPHYSICAL TECHNIQUES OF FETAL EVALUATION

Descriptions of the predictive value of several commonly performed antenatal tests of fetal well-being are presented in Table 27.2, and details on each methodology are given in the sections that follow.

Maternal Assessment of Fetal Activity

Studies performed using real-time ultrasonography have demonstrated that during the third trimester, the human fetus spends 10% of its time making gross fetal body movements and that 30 such movements are

TABLE 27.2 Comparison of Selected Antenatal Tests		
Test	False-Negative Rate (%)	False-Positive Rate (%)
Contraction stress test	0.04	35–65
Nonstress test	0.2–0.8	55–90
Biophysical profile	0.07–0.08	40–50
Modified biophysical profile	0.08	60

Modified from Signore C, Freeman RK, Spong CY. Antenatal testing: a reevaluation. Executive Summary of a Eunice Kennedy Shriver National Institute of Child Health and Human Development Workshop. *Obstet Gynecol.* 2009;113:687-701.

made each hour.⁷⁴ Periods of active fetal body movement last about 40 minutes, whereas quiet periods last about 20 minutes. Patrick and colleagues⁷⁴ noted that the longest period without fetal movements in a normal fetus was about 75 minutes. The woman is able to perceive about 70% to 80% of gross fetal movements. The fetus does make fine body movements such as limb flexion and extension, hand grasping, and sucking; these probably reflect more coordinated CNS function, although the woman is generally unable to perceive these fine movements. Fetal movement appears to peak between 9:00 PM and 1:00 AM, a time when maternal glucose levels are falling.^{71,74} In a study in which maternal glucose levels were carefully controlled with an artificial pancreas, Holden and coworkers found that hypoglycemia was associated with increased fetal movement. Fetal activity does not increase after meals or after maternal glucose administration.⁷⁵

The decrease in fetal movement with hypoxemia makes maternal assessment of fetal activity a potentially simple and widely applicable method of monitoring fetal well-being. However, prospective trials of this method for prevention of perinatal mortality have failed to conclusively show benefit.² Neldam demonstrated a 73% reduction in avoidable fetal deaths in a prospective trial of more than 1500 women instructed to count fetal movements. In contrast, a subsequent international trial⁷⁶ of more than 68,000 women randomized to routine fetal movement assessment versus no formal assessment of fetal activity failed to show a significant reduction in fetal deaths in the group randomized to routine movement counting. This was true in the latter, compared with the former, trial and throughout the literature on this topic. This illustrates the difficulties in validating and reproducing the results of these trials and the uncertain clinical benefit that may be derived from introducing maternal assessment of fetal movement into routine clinical practice. A 2015 Cochrane review⁷⁷ of five trials that involved more than 71,000 pregnancies concluded that evidence is insufficient to recommend routine fetal movement counting to prevent fetal death.

Despite these results, this type of fetal assessment may offer some advantages. Although the range in fetal activity will be wide but normal, with FMC, each woman and her fetus serve as their own control. Factors that affect maternal perception of fetal movement are not well understood. Fetal and placental factors that may contribute include placental location, the length and type of fetal movements, and AFV, although whether AFV affects maternal perception or actual fetal movement is unclear.⁷² Maternal factors that may influence the evaluation of fetal movement include maternal activity, parity, obesity, medications, and psychological factors, including anxiety. Studies of these associations have demonstrated conflicting results.⁷²

Several methods have been used to monitor fetal activity in research and clinical practice. These methods include FMC over a prescribed time period, such as 30 to 60 minutes one to three times daily, or conversely a target number of fetal movements to be counted over a variable time range. A variety of “normal” and “abnormal” FMC results or thresholds have been proposed, to which the patient should be instructed to respond by presenting for further evaluation of the condition of the fetus.

Potential negative impacts of maternal assessment of fetal activity include maternal anxiety and a potential increase in utility of other testing modalities and antepartum admissions. However, given the likely low impact of false-positive FMC results, as well as the fact that this has been the only form of antenatal testing to be compared with no testing and found to have a favorable impact on fetal death, it seems reasonable to continue to prescribe some form of maternal

assessment of fetal movement to patients until more conclusive evidence can be generated. As stated by Froen and associates⁷⁸ in a 2008 review, “The fact that FMC versus no counting seems to be beneficial irrespective of the chosen definition of decreased fetal movement (DFM) would support the hypothesis that benefit was derived from increased maternal vigilance regarding FM. When awareness, vigilance, and fetal movement counting is to be promoted, we have suggested that a maternal perception of significant and sustained reduction in fetal activity should remain the main definition of DFM. Any alarm limits, as ‘10 FM in 2 hours,’ should only be for guidance as a ‘rule of thumb.’ ”

Given the findings from population-based studies of potentially preventable stillbirths, as mentioned earlier, it may be that the greatest barriers to the use of FMC as a primary effective method of antenatal surveillance are the lack of knowledge of a best practice response by patients and health care providers to the complaint of DFM.⁷⁹ In both research and clinical settings, significant differences persist in the method of FMC, the definition of “abnormal” fetal movements, patient compliance with the intervention, and provider response to patients who present to care as a result of an abnormal FMC. Effects of this inconsistency were illustrated in a 2016 meta-analysis of 23 publications, including three randomized trials, demonstrating no consistent impact of interventions to enhance awareness of DFM on rates of fetal death.⁷⁹

Contraction Stress Test

The *contraction stress test* (CST), also known as the *oxytocin challenge test* (OCT), was the first biophysical technique widely applied for antepartum fetal surveillance. It was well known that uterine contractions produced a reduction in blood flow to the intervillous space. Analyses of intrapartum fetal heart rate monitoring had shown that a fetus with inadequate placental respiratory reserve would demonstrate recurrent late decelerations in response to hypoxia, in particular a fetal arterial oxygen pressure below 20 mm Hg (see [Chapter 17](#)). As noted earlier, this drop in fetal heart rate in response to transient hypoxia is mediated by a vagal response to transient systemic fetal vascular reactivity provoked by hypoxia. The CST extended these observations to the antepartum period. The response of the fetus at risk for uteroplacental insufficiency to uterine contractions formed the basis for this test.

To perform the CST, the patient is placed in the semi-Fowler position at a 30- to 45-degree angle with a slight left tilt to avoid the supine hypotensive syndrome. Continuous external fetal heart rate and uterine contraction monitoring is recorded, and a baseline is obtained before stimulating uterine activity. Maternal blood pressure is determined every 5 to 10 minutes to detect maternal hypotension. In some cases, adequate uterine activity occurs spontaneously and additional uterine stimulation is unnecessary. An adequate CST requires uterine contractions of moderate intensity that last about 40 to 60 seconds with a frequency of three in 10 minutes. These criteria were selected to approximate the stress experienced by the fetus during the first stage of labor. If uterine activity is absent or inadequate, intravenous oxytocin is begun to initiate contractions, and it is increased until adequate uterine contractions have been achieved.²¹ Several methods of nipple stimulation have been used to induce adequate uterine activity, and the success rate at achieving adequate contractions and test results is comparable to that of oxytocin infusion.²¹ After the CST has been completed, the patient should be observed until uterine activity has returned to its baseline level.

Contraindications to the test include a high risk for premature labor, such as in patients with PPROM, multiple gestations, and cervical incompetence, although the CST has not been associated with an increased incidence of premature labor.⁸⁰ The CST should also be avoided in conditions in which uterine contractions may be dangerous, such as with placenta previa or a previous classic cesarean delivery or uterine surgery.

Most clinicians use the definitions proposed by Braly and Freeman and colleagues^{80,81} to interpret the CST (detail in Table 27.3):

- Negative: No late or significant variable decelerations
- Positive: Late decelerations with at least 50% of contractions
- Suspicious: Intermittent late or variable decelerations
- Hyperstimulation: Decelerations with contractions longer than 90 seconds' duration or a greater than 2-minute frequency
- Unsatisfactory: Fewer than three contractions per 10 minutes or an uninterpretable tracing

Predictive Value of the Contraction Stress Test

A negative CST has been consistently associated with good fetal outcome. Studies have shown the incidence of perinatal death within 1 week of a negative CST (i.e., the false-negative rate) to be less than 1 per 1000.^{2,82,83} Many of these deaths, however, can be attributed to cord accidents, malformations, placental abruption, and acute deterioration of glucose control in patients with diabetes. Thus the CST, like most methods of antepartum fetal surveillance, cannot predict acute fetal compromise. If the CST is negative and reactive, a repeat study is

usually scheduled in 1 week (Fig. 27.9). A negative but nonreactive CST is not suggestive of acute fetal compromise but cannot be seen as constituting the same low false-negative rate over the course of 1 week as a negative reactive CST. A negative nonreactive CST is usually repeated in 24 hours (Fig. 27.10). Changes in the patient's clinical condition may warrant more frequent studies. A positive CST has been associated with an increased incidence of intrauterine death, late decelerations in labor, low 5-minute Apgar scores, IUGR, and meconium-stained amniotic fluid (Fig. 27.11)⁸² with an overall likelihood of perinatal death after a positive CST ranging from 7% to 15%. On the other hand, the incidence of false-positive CSTs has been significant; depending on the end point used, it can be between 35% and 65%.^{2,83} The positive CST is more likely to be associated with fetal compromise if the baseline heart rate lacks accelerations and the latency period between the onset of the uterine contractions and the onset of the late deceleration is less than 45 seconds.

The high incidence of false-positive CSTs is one of the greatest limitations of this test because such results could lead to unnecessary premature intervention. False-positive CSTs may be attributable to misinterpretation of the tracing; supine hypotension, which decreases uterine perfusion; uterine hyperstimulation, which is not appreciated using the tocodynamometer; or an improvement in fetal condition after the CST has been performed. The high false-positive rate also indicates that a patient with a positive CST need not necessarily require an elective cesarean delivery. A suspicious or equivocal CST or one that is unsatisfactory or shows hyperstimulation should be

TABLE 27.3 Interpretation of Contraction Stress Test		
Finding	Description	Incidence (%)
Negative	No late decelerations appear anywhere on the tracing with adequate uterine contractions (3 in 10 min).	80
Positive	Late decelerations that are consistently and persistently present with the majority (>50%) of contractions without excessive uterine activity; if persistent late decelerations are seen before the frequency of contractions is adequate, the test is interpreted as positive.	3–5
Suspicious	Decelerations are late and inconsistent.	5
Hyperstimulation	Uterine contractions closer than every 2 min or lasting more than 90 s or five uterine contractions in 10 min; if no late decelerations are seen, the test is interpreted as negative.	5
Unsatisfactory	The quality of the tracing is inadequate for interpretation or adequate uterine activity cannot be achieved.	5



Fig. 27.9 Reactive and Negative Contraction Stress Test (CST). Negative CST. No fetal heart rate decelerations following any of the three contractions lasting 40 to 60 seconds over a 10-minute period. Fetal heart rate is reactive.

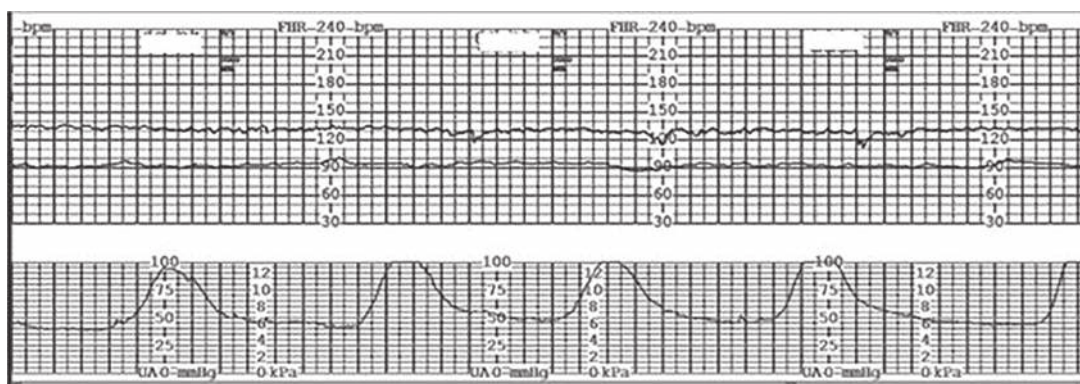


Fig. 27.10 Nonreactive and Negative Contraction Stress Test. No significant fetal heart rate decelerations following any of the four contractions lasting 40 to 60 seconds over a 10-minute period. Fetal heart rate is nonreactive.

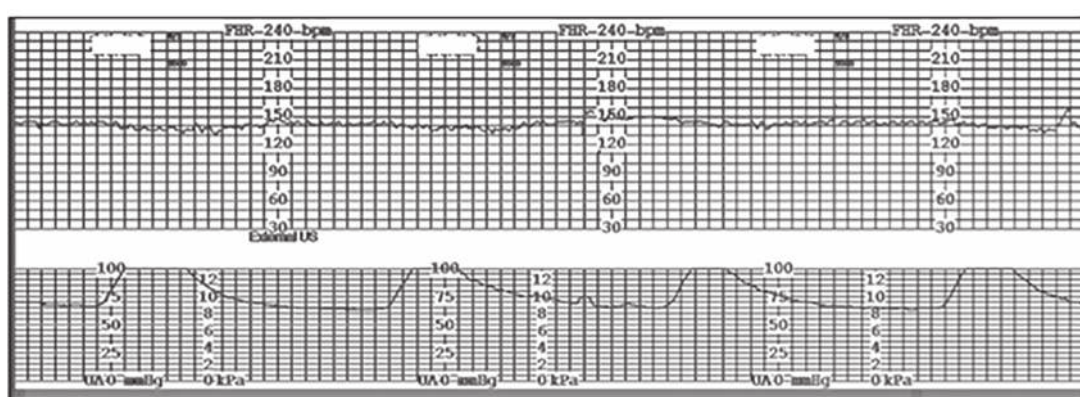


Fig. 27.11 Nonreactive and Positive Contraction Stress Test. Subtle late fetal heart rate decelerations follow the first two of the four contractions. Fetal heart rate is nonreactive.

repeated in 24 hours. In one series, 7.5% of patients with an initially suspicious CST exhibited positive tests on further evaluation, 53.7% became negative, and 38.8% remained suspicious.⁸⁴

While the CST has stood the test of time in terms of reliability and predictive value, it rarely has a routine place in modern obstetric practice. The cost and cumbersome nature of implementation, including its inconvenience for patients, has limited its role.

Nonstress Test

Observations made in the mid-20th century that accelerations of the fetal heart rate in response to fetal activity, uterine contractions, or stimulation reflect fetal well-being formed the basis for the NST. The NST is the most widely applied technique for antepartum fetal evaluation despite the uncertainty regarding its reliability and reproducibility as a test of fetal assessment. The basic technology that underlies its performance and application has changed little since its introduction and wide acceptance into antenatal practice.⁸⁵

In late gestation, the healthy fetus exhibits an average of 34 accelerations above the baseline fetal heart rate each hour.⁸⁶ These accelerations, which average 20 to 25 beats/min in amplitude and about 40 seconds in duration, require intact neurologic coupling between the fetal CNS and the fetal heart.⁸⁶ Fetal hypoxia disrupts this pathway. At term, fetal accelerations are associated with fetal movement more than 85% of

the time, and more than 90% of gross movements are accompanied by accelerations. Fetal heart rate accelerations may be absent during periods of quiet fetal sleep. Studies by Patrick and associates⁸⁶ demonstrated that the longest time between successive accelerations in the healthy term fetus is about 40 minutes. However, the fetus may fail to exhibit heart rate accelerations for up to 80 minutes and still be normal.

Although an absence of fetal heart rate accelerations is most often attributable to a quiet fetal sleep state, CNS depressants such as narcotics and phenobarbital, as well as the β -blocker propranolol, can reduce heart rate reactivity. Chronic smoking is known to decrease fetal oxygenation through an increase in fetal carboxyhemoglobin and a decrease in uterine blood flow. Fetal heart rate accelerations are also decreased in smokers.

The NST is usually performed in an outpatient setting. In most cases, only 10 to 15 minutes are required to complete the test. It has virtually no contraindications, and few equivocal test results are observed. The patient may be seated in a reclining chair, with care taken to ensure that they are tilted to the left to avoid the supine hypotensive syndrome.²¹ The patient's blood pressure should be recorded before the test is begun and then repeated at 5- to 10-minute intervals. The fetal heart rate is monitored using the Doppler ultrasound transducer, and the tocodynamometer is applied to detect uterine contractions or fetal movement. Fetal activity may be recorded

by the patient using an event marker, or it may be noted by the staff performing the test.

Fetal Heart Rate Patterns Observable on the Nonstress Test

Reactive. The most widely applied definition of a reactive test requires that at least two accelerations of the fetal heart rate, each with a peak amplitude of 15 beats/min and total duration of 15 seconds, be observed in 20 minutes of monitoring (Fig. 27.12).⁸⁷ Gestational age affects reactivity criteria because sympathetic and parasympathetic influences on fetal heart rate change with advancing gestational age. Compared with the early third trimester, increased frequency and amplitude of fetal heart rate accelerations are seen after 30 weeks' gestation; although half of normal fetuses demonstrate accelerations with fetal movements at 24 weeks, nearly all fetuses will do so after 30 weeks. Similarly, before 32 weeks' gestation, acceptable criteria for a reactive fetal heart rate tracing include accelerations with a peak of only 10 beats/min amplitude and 10 seconds' duration rather than the widely applied 15-and-15 definition for reactivity described earlier.⁸⁷

Overall, on initial testing, 85% of NSTs are reactive and 15% are nonreactive.⁸⁸ Fewer than 1% of NSTs prove unsatisfactory because of inadequately recorded fetal heart rate data.

Nonreactive. If the criteria for reactivity are not met, the test is considered nonreactive (Fig. 27.13). The most common cause for a nonreactive test is a period of fetal inactivity or quiet sleep. Therefore the test may be extended for an additional 20 minutes with the expectation that the fetal state will change and reactivity will appear. In an effort to change the fetal state, some clinicians have manually stimulated the fetus or attempted to increase fetal glucose levels by giving the woman orange juice, but no evidence suggests that such efforts will increase fetal activity. If the test has been extended for

40 minutes, and reactivity has not been seen, a BPP or CST should be performed. **Most fetuses who exhibit a nonreactive NST are not compromised but simply fail to exhibit heart rate reactivity during the 40-minute period of testing.** Of those fetuses who exhibit a nonreactive NST, about 25% will have a positive CST on further evaluation.⁸⁸ Reactivity that occurs during preparations for the CST has proved to be a reliable index of fetal well-being.

The likelihood of a nonreactive test is substantially increased with early gestational age. Between 24 and 28 weeks' gestation, about 50% of NSTs are nonreactive, and 15% of NSTs remain nonreactive between 28 and 32 weeks.⁸⁹ After 32 weeks, the incidences of reactive and nonreactive test results are comparable with those seen at term. Malformed fetuses also exhibit a significantly higher incidence of nonreactive NSTs.⁹⁰

Vibroacoustic stimulation (VAS) may be used to change the fetal state from quiet to active sleep (Fig. 27.14). Auditory brainstem response appears to be functional in the fetus at 26 to 28 weeks' gestation. Thus VAS may significantly increase the incidence of reactive NSTs after 26 weeks' gestation and may reduce the testing time, making this a potentially useful adjunct to antepartum fetal testing.⁸⁹ Most studies of VAS have used an electronic artificial larynx that generates sound pressure levels measured at 1 m in air of 82 dB with a frequency of 80 Hz and a harmonic of 20 to 9000 Hz. Whether it is the acoustic or vibratory component of this stimulus that alters fetal state is unclear. VAS may produce a significant increase in the mean duration and amplitude of heart rate accelerations, fetal heart rate variability, and gross fetal body movements within 3 minutes of stimulation.⁹¹ Several studies using VAS have shown a decreased incidence of nonreactive NSTs from 13% to 14% down to 6% to 9%. A reactive NST after VAS stimulation appears to be as reliable an index of fetal well-being as spontaneous reactivity.⁹¹ However, those fetuses

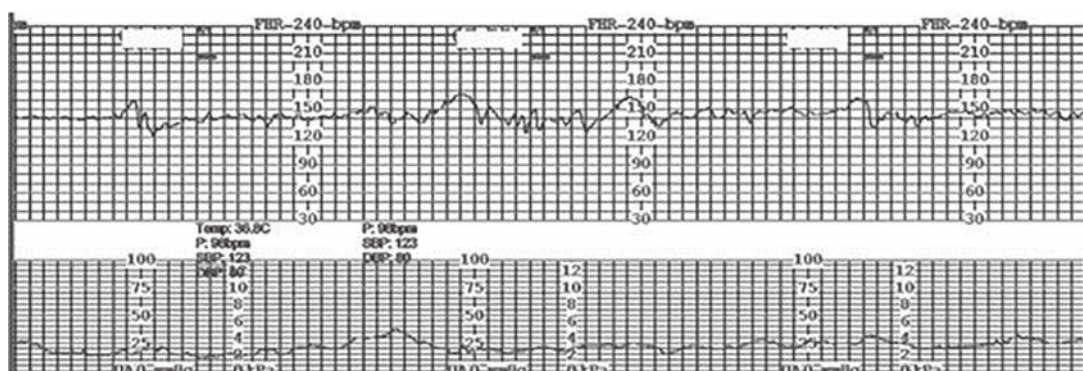


Fig. 27.12 Reactive Nonstress Test. Fetal heart rate baseline is 140 bpm with several 15 × 15 accelerations present.

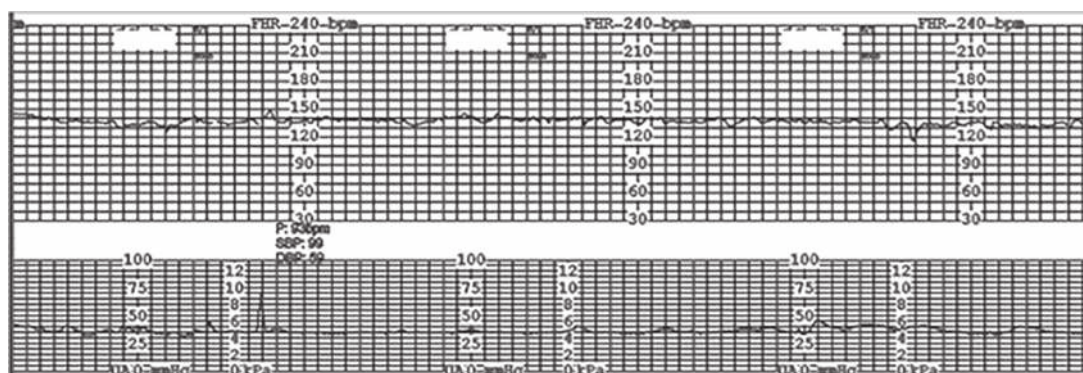


Fig. 27.13 Nonreactive Nonstress Test. No 15 × 15 accelerations of the fetal heart rate are observed.

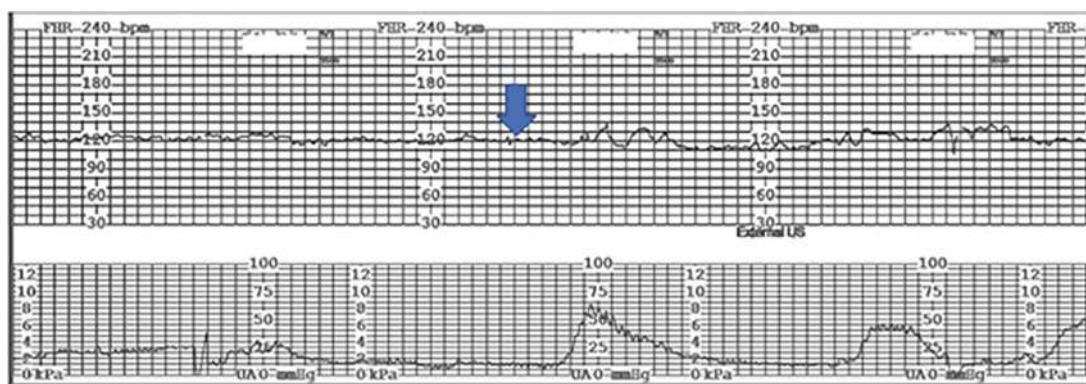


Fig. 27.14 Reactive Nonstress Test After Vibroacoustic Stimulation. Fetal heart rate is without 15 × 15 accelerations present until following VAS (blue arrow).

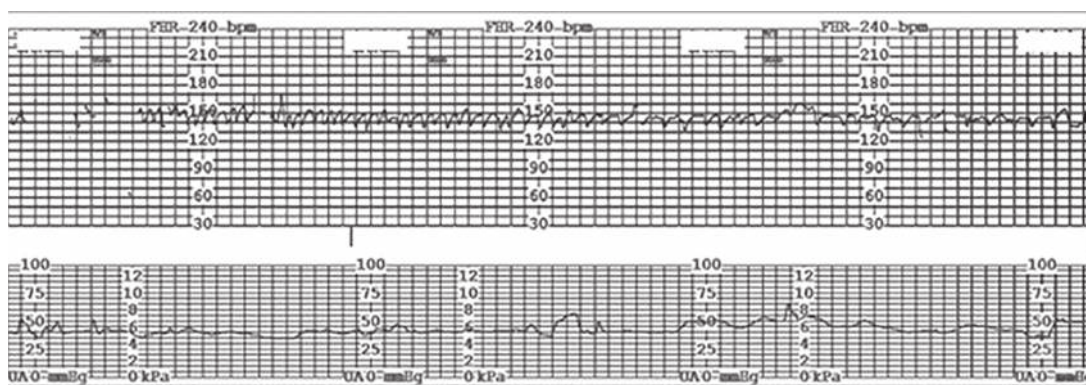


Fig. 27.15 Sinusoidal Fetal Heart Rate Pattern. Baseline is stable at 145 bpm with regular oscillations of 5 to 10 bpm without accelerations.

who remain nonreactive even after VAS may be at increased risk for poor perinatal outcome, with increased rates of intrapartum fetal distress and growth restriction and low Apgar scores.

In most centers that use VAS, the baseline fetal heart rate is first observed for 10 to 20 minutes. If the pattern is nonreactive, a stimulus of 3 seconds or less is applied near the fetal head. If the NST remains nonreactive, the stimulus is repeated at 1-minute intervals up to three times. If the fetus still does not respond, further evaluation should be carried out with a BPP or CST. Studies have confirmed the safety of VAS use during pregnancy, and no long-term evidence of hearing loss has been found in children followed in the neonatal period and up to 4 years of age.⁹² Other interventions to safely shorten NST time have been investigated, most with equivocal results.⁹³ **In summary, VAS may be helpful in shortening the time required to perform an NST and may be especially useful in centers where large numbers of NSTs are done.**

Other Nonstress Test Patterns or Findings

Sinusoidal Pattern. On rare occasions, a sinusoidal heart rate pattern may be observed, as described in Chapter 17. This undulating heart rate pattern with virtually absent variability (Fig. 27.15) has been associated with fetal anemia and asphyxia, congenital malformations, and medications such as narcotics. In one of the earliest reports on the use of the NST, Rochard and coworkers⁹⁴ described a sinusoidal pattern in 20 of 50 pregnancies complicated by Rh isoimmunization. One-half of these pregnancies ended in a perinatal death, and 40% of the surviving infants required prolonged hospitalization. Only 10% of the babies with a sinusoidal pattern had an uncomplicated course.

Bradycardia. Before 27 weeks' gestation, the normal fetal heart rate response to fetal movement may, in fact, be a bradycardia. However, in settings with a high a priori risk of fetal compromise, such as IUGR associated with antiphospholipid syndrome, bradycardia at a gestational age of 26 to 28 weeks may be a predictor of impending fetal death.

Significant antepartum fetal heart rate bradycardias have been observed in 1% to 2% of all NSTs, defined as a fetal heart rate of 90 beats/min or a fall in the fetal heart rate of 40 beats/min below the baseline for 60 seconds or longer (Fig. 27.16). Note that this terminology is distinct from the intrapartum fetal heart rate tracing nomenclature categorizing a deceleration lasting greater than 2 but fewer than 10 minutes as a "prolonged deceleration" (Chapter 17). In a review of 121 cases, antepartum bradycardia was associated with increased perinatal morbidity and mortality, particularly antepartum fetal death, cord compression, IUGR, and fetal malformations.⁹⁵ Although about one-half of the NSTs associated with bradycardia were reactive, the incidence of a nonreassuring fetal heart rate pattern in labor that led to emergency delivery in this group was identical to that of patients who exhibited nonreactive NSTs. Clinical management decisions should be based on the finding of bradycardia, *not* on the presence or absence of reactivity. Bradycardia has a higher positive predictive value for fetal compromise (fetal death or fetal intolerance of labor) than does the nonreactive NST. **Expectant management in the setting of a bradycardia has been associated with a PMR of 25%.** Therefore preparations for delivery should be undertaken after careful consideration of gestational age–related risks and appropriate counseling.

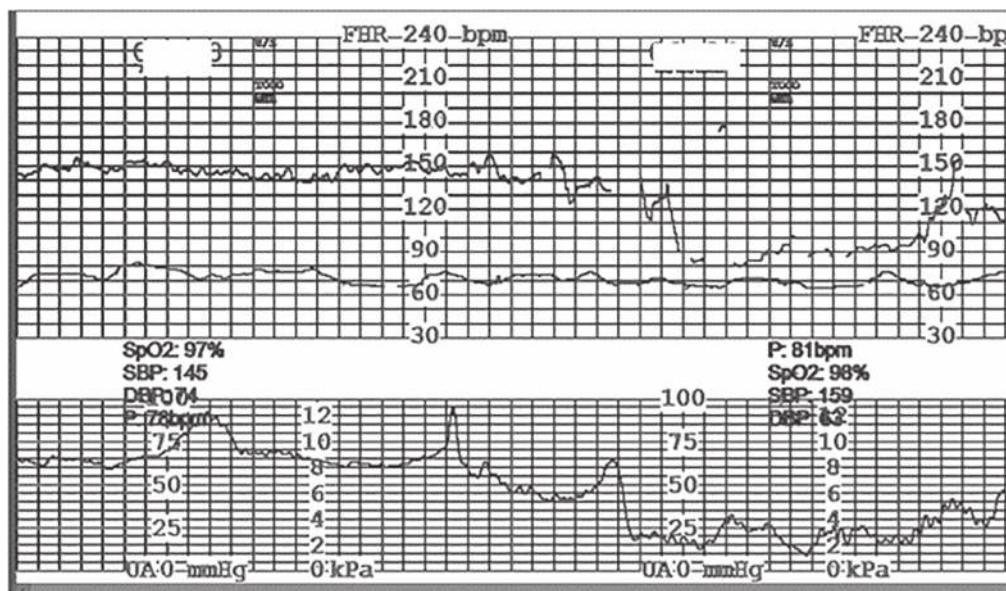


Fig. 27.16 Spontaneous Bradycardia. Fetal heart rate decelerates to 80 bpm without recovery to baseline in this panel.

Tachycardia. The fetal heart rate baseline evaluation must also take gestational age into account, as with fetal heart rate reactivity. Vagal activity has a greater influence on fetal heart rate at baseline as gestation advances; thus baseline fetal heart rate will decrease from an average of 155 beats/min at 20 weeks to 145 beats/min at 30 weeks. **The most common etiology of fetal tachycardia is maternal-fetal fever secondary to maternal-fetal infection such as chorioamnionitis.** Other causes include chronic hypoxemia, maternal hyperthyroidism, maternal medication exposure, and fetal tachyarrhythmia. Fetal heart rates above 200 beats/min, and certainly above 220 beats/min, should increase the index of suspicion of fetal tachyarrhythmia and lead to further fetal cardiac evaluation with a targeted fetal echocardiogram. For fetal heart rates between 160 and 180 beats/min, the presence or absence of baseline variability is an important indicator of fetal acid-base status. Fetal acidosis is more likely if baseline variability is absent.⁸⁷

Arrhythmia. Among fetal arrhythmias, most (approximately 90%) are tachyarrhythmias. Fetal tachyarrhythmia is most often diagnosed when the fetal ventricular heart rate is greater than 180 beats/min, but usually greater than 200 beats/min. Common causes of fetal tachyarrhythmias are paroxysmal supraventricular tachycardia and atrial flutter. The administration of antiarrhythmic therapy to pregnant individuals carrying fetuses with sustained supraventricular tachycardia represented the first examples of successful prenatal cardiac therapy. Multiple publications describe treatment protocols for this arrhythmia. Characteristics of fetuses who may need treatment include hydrops fetalis in the face of a sustained arrhythmia and a gestational age too early to permit safe delivery and postnatal treatment. In such cases, therapy is best initiated with medications that have a relatively broad therapeutic margin and a low risk for proarrhythmia (unwanted precipitation or exacerbation of the arrhythmia) for the fetus and pregnant individual. Digoxin is the most commonly used first-line agent but is rarely effective in the presence of fetal hydrops, for which sotalol or flecainide are typical first-line choices.⁹⁶

Fetal bradyarrhythmia is diagnosed when the fetal ventricular heart rate is slower than 100 beats/min, usually secondary to atrioventricular block. About half of all cases are caused by fetal cardiac anomalies, and of the remaining cases with normal cardiac structure, many are related

to maternal antibodies. These antibodies, which include SS-A and SS-B, may directly target the fetal atrioventricular node or the myocardium, resulting in heart block and myocarditis. Fetuses with bradyarrhythmia may develop hydrops fetalis, particularly with sustained bradycardia at a rate of less than 55 beats/min. In utero heart failure with congenital heart block, with or without congenital heart disease, represents an indication for electrical pacemaker therapy in surviving neonates. This condition is less often successfully treated antenatally than fetal tachyarrhythmia. Available therapies include maternal administration of β -agonists, which have been shown to increase fetal ventricular rate by 10% to 20%, and steroid or immunoglobulin administration, which may be effective in cases of fetal heart block caused by maternal antibodies. Potential improved fetal outcome includes reversal of hydrops in several reports.⁹⁶

Deceleration. In most cases, mild variable decelerations are not associated with poor perinatal outcome. Meis and associates⁹⁷ reported that variable decelerations of 20 beats/min or more below the baseline heart rate but lasting less than 10 seconds were noted in 50.7% of patients undergoing an NST (Fig. 27.17). Whereas these decelerations were more often associated with a nuchal cord, they were not predictive of IUGR, a nonreassuring fetal heart rate pattern, or more severe variable decelerations during labor. When mild variable decelerations are observed, even if the NST is reactive, an ultrasound examination should be performed to rule out oligohydramnios. A low AFV and mild variable decelerations increase the likelihood of a cord accident. If late decelerations in response to contractions are observed in the performance of an NST, criteria for interpretation of a CST should then be used.

Predictive Value of the Nonstress Test

The NST is most predictive when it is normal or reactive. The reported false-negative rate over multiple studies ranges from 0.2% to 0.8%, which corresponds to a fetal death rate of 3 to 8 per 1000 within 1 week of a reactive NST. The false-positive rate is considerably higher and ranges from 50% to more than 90% in various studies.²

A 2015 Cochrane review⁸⁵ integrated the results of six randomized controlled trials that included 2105 pregnancies; four studies with 1636

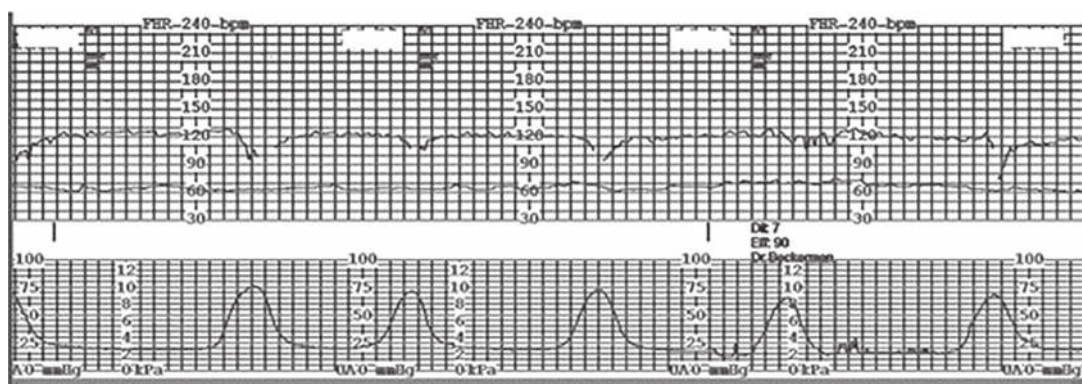


Fig. 27.17 Variable Decelerations. Fetal heart rate with abrupt decrease and quick return to baseline. These variable decelerations are occurring with contractions.

pregnant women compared nonstress testing (or NST with results revealed) to no NST (or NST with results concealed), and two studies with 469 pregnant women compared NST with computerized analysis to traditional (visually analyzed) NST. The six included studies recruited only high-risk pregnant women, and none provided information about singleton and multiple pregnancies. Of note, all four of the studies that compared NST with no NST were performed in the 1980s. In the studies that compared NST with no NST, no significant differences were identified in the risk for perinatal mortality (RR, 2.05; 95% CI, 0.95 to 4.42; 2.3% vs. 1.1%). In addition, no differences were identified in rates of cesarean delivery, “potentially preventable perinatal mortality,” Apgar scores, neonatal intensive care unit (NICU) admissions, gestational age at birth, or neonatal seizures. However, in the two studies that compared computerized analysis to visual analysis, a significant reduction was seen in perinatal mortality with computerized analysis (RR, 0.20; 95% CI, 0.04 to 0.88; 0.9% vs. 4.2%).⁸⁵ The authors concluded that the analysis was underpowered to detect possible important differences in perinatal mortality and pointed out that many aspects of antenatal and postnatal care may have changed since the included trials were performed.

The authors of the previously mentioned review also called for new studies to assess the effects of the traditional and computer-analyzed NST to assess the true impact on perinatal mortality and other outcomes. At present, computerized NST analysis techniques are still considered investigational and are not widely implemented in current clinical practice. This is despite the fact that one of the pitfalls of traditional NST analysis has historically been interobserver variability, or operator dependence of those clinicians interpreting the NST. Both interobserver and intraobserver variability have been demonstrated to be as high as 0.5.⁹⁸

In selected high-risk pregnancies, the false-negative rate associated with a weekly NST may be unacceptably high, especially in those pregnancies complicated by pregestational diabetes mellitus, FGR, and prolonged gestation. In these cases, increasing the frequency of the NST to twice weekly may be advisable.⁹⁹

Fetal Biophysical Profile

The use of real-time ultrasonography to assess antepartum fetal condition has enabled the obstetrician to perform an *in utero* physical examination and evaluate dynamic functions that reflect the integrity of the fetal CNS. As emphasized by Manning and colleagues,¹⁰⁰ “fetal biophysical scoring rests on the principle that the more complete the examination of the fetus, its activities, and its

environment, the more accurate may be the differentiation of fetal health from disease states.”

Fetal breathing movements (FBMs) were the first biophysical parameter to be assessed using real-time ultrasonography. It is thought that the fetus exercises its breathing muscles *in utero* in preparation for postdelivery respiratory function. With real-time ultrasonography, FBM is evidenced by downward movement of the diaphragm and abdominal contents and by an inward collapsing of the chest. FBMs become regular at 20 to 21 weeks and are controlled by centers on the ventral surface of the fourth ventricle of the fetus.¹⁰¹ They are observed about 30% of the time, are seen more often during REM sleep, and demonstrate intact neurologic control when present. Although the absence of FBMs may reflect fetal asphyxia, this finding may also indicate that the fetus is in a period of quiet sleep.⁷¹ Several factors other than fetal state and hypoxia can influence the presence of FBM. As maternal glucose levels rise, FBM becomes more frequent, and during periods of maternal hypoglycemia, FBM decreases. Maternal smoking reduces FBM, probably as a result of fetal hypoxemia. Narcotics that depress the fetal CNS also decrease FBM.

Regarding other observable fetal states as markers of fetal oxygen status and well-being, Vintzileos and coworkers¹⁰¹ noted that **fetal biophysical activities that are present earliest in fetal development are the last to disappear with fetal hypoxia.** The fetal tone center in the cortex begins to function at 7.5 to 8.5 weeks, therefore fetal tone would be the last fetal parameter to be lost with worsening fetal condition. The fetal movement center in the cortex nuclei is functional at 9 weeks and would be more sensitive than fetal tone. As noted earlier, FBM becomes regular at 20 to 21 weeks. Finally, fetal heart rate control, which resides within the posterior hypothalamus and medulla, becomes functional at the end of the second trimester and early in the third trimester. An alteration in fetal heart rate would theoretically be the earliest sign of fetal compromise.

Using these principles, Manning and colleagues¹⁰² developed the concept of the fetal BPP score. These researchers elected to combine the NST with four parameters that could be assessed using real-time ultrasonography: FBM, fetal movement, fetal tone, and AFV (Videos 27.1 to 27.4 and Figs. 27.18 to 27.20; see Chapter 28). FBM, fetal movement, and fetal tone are mediated by complex neurologic pathways and should reflect the function of the fetal CNS at the time of the examination. On the other hand, AFV should provide information about the presence of chronic fetal asphyxia. Finally, the ultrasound examination performed for the BPP has the added advantage of detecting previously unrecognized major fetal anomalies. A BPP score

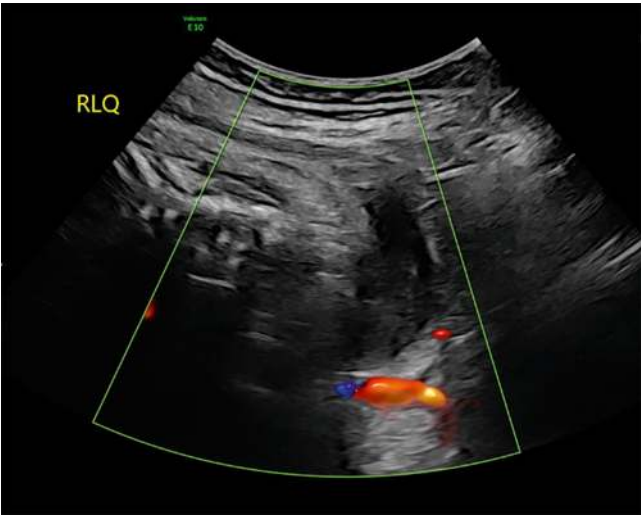


Fig. 27.18 Amniotic Fluid Volume. To obtain the amniotic fluid index, the maximum cord-free vertical pocket of amniotic fluid in each of the four quadrants of the uterus is measured and summed.

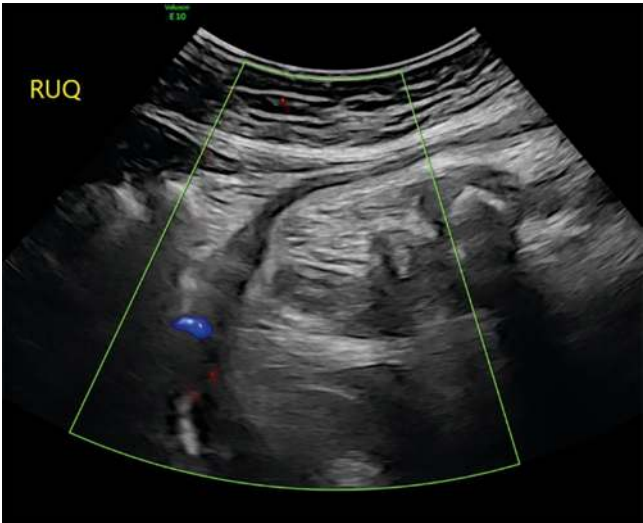


Fig. 27.19 Amniotic Fluid Volume. Sometimes, no cord-free pocket of fluid is found, as in the right upper quadrant (RUQ) of this patient.

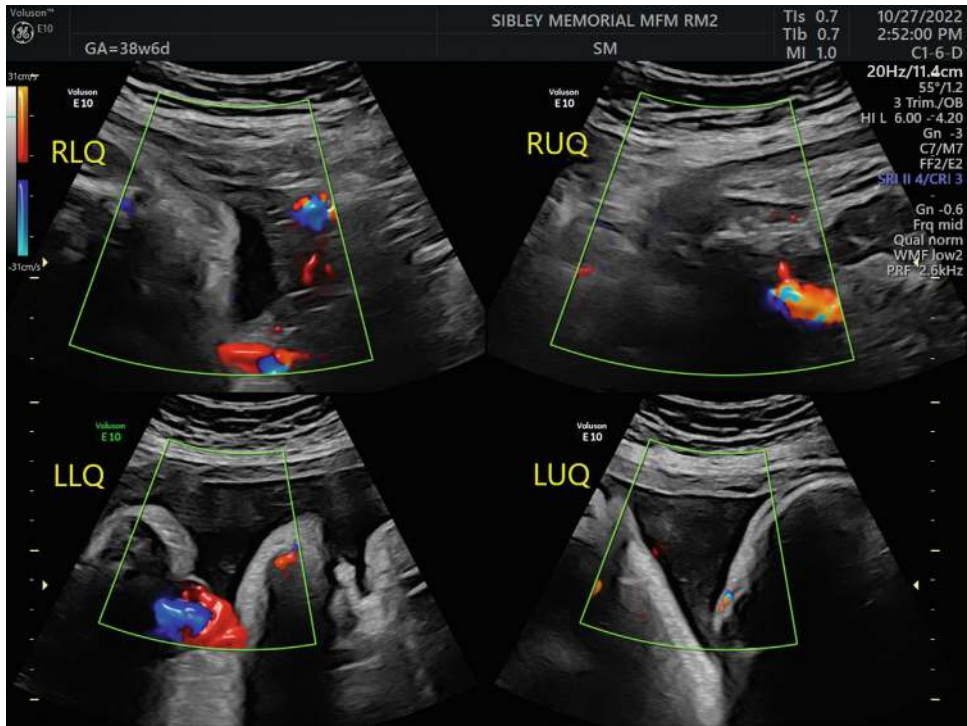


Fig. 27.20 Amniotic Fluid Index. Four quadrant maximum vertical pocket of amniotic fluid.

that is similar to the Apgar score used to assess the condition of the newborn was developed.¹⁰² The presence of a normal parameter, such as a reactive NST, was awarded two points, whereas the absence of that parameter was scored as 0. The highest score a fetus can receive is 10, and the lowest score is 0. The BPP may be used as early as 26 to 28 weeks' gestation. The time required for the fetus to achieve a satisfactory BPP score is closely related to fetal state, with an average of only 5 minutes if the fetus is in a 2F state but over 25 minutes if it is in a 1F state.

The criteria proposed by Manning and colleagues¹⁰² are shown in Table 27.4, and the clinical actions recommended in response to these scores are presented in Table 27.5. A low score on the BPP does not prohibit attempted vaginal delivery if other obstetric factors are favorable. In a prospective blinded study of 216 high-risk patients, no perinatal deaths were observed when all five variables described earlier were normal, but a PMR of 60% was seen in fetuses with a score of zero.¹⁰² Fetal deaths were increased 14-fold with the absence of fetal movement, and the PMR was increased 18-fold if FBM was absent. Any single test

TABLE 27.4 Criteria of Biophysical Profile Scoring

Biophysical Variable	Normal (Score = 2)	Abnormal (Score = 0)
Fetal breathing movements	At least 1 episode of ≥ 30 sec duration in 30-min observation	Absent or no episode of ≥ 30 sec duration in 30 min
Gross body/limb movement	At least 3 discrete body/limb movements in 30 min (episodes of active continuous movement considered a single movement)	Up to 2 episodes of movements in 30 min
Fetal tone	At least 1 episode of active extension with return-flexion of fetal limb or trunk, with opening and closing of the hand considered to reflect normal tone	Either slow extension with return-partial flexion, movement of limb in full extension, or absent fetal movement
Reactive fetal heart rate	At least 2 episodes of acceleration of ≥ 15 beats/min and 15 sec duration associated with fetal movement in 20 min ^a	Fewer than 2 accelerations or acceleration < 15 beats/min in 20 min
Amniotic fluid volume	At least 1 pocket of amniotic fluid measuring ≥ 2 cm in 2 perpendicular planes	Either no amniotic fluid pockets or a pocket < 2 cm in 2 perpendicular planes

^aFor gestational age > 30 weeks.

Modified from Manning FA. Biophysical profile scoring. In: Nijhuis J, ed. *Fetal Behaviour*. New York: Oxford University Press; 1992:241.

TABLE 27.5 Management Based on Biophysical Profile

Score	Interpretation	Management
10	Normal; low risk for chronic asphyxia	Repeat testing at weekly to twice-weekly intervals.
8	Normal; low risk for chronic asphyxia	Repeat testing at weekly to twice-weekly intervals.
6	Suspect chronic asphyxia	If ≥ 36 –37 weeks' gestation or < 36 weeks' gestation with positive testing for fetal pulmonary maturity, consider delivery; if < 36 weeks' gestation and/or fetal pulmonary maturity testing is negative, repeat biophysical profile in 4–6 h; deliver if oligohydramnios is present.
4	Suspect chronic asphyxia	If ≥ 36 weeks' gestation, deliver; if < 32 weeks' gestation, repeat score.
0–2	Strongly suspect chronic asphyxia	Extend testing time to 120 min; if persistent score is 4 or less, deliver regardless of gestational age.

Modified from Manning FA, Harman CR, Morrison I, et al. Fetal assessment based on fetal biophysical profile scoring. *Am J Obstet Gynecol*. 1990;162:703; and Manning FA. Biophysical profile scoring. In: Nijhuis J, ed. *Fetal Behaviour*. New York: Oxford University Press; 1992:241.

was associated with a significant false-positive rate that ranged from 50% to 79%. However, combining abnormal variables significantly decreased the false-positive rate to as low as 20%. The false-negative rate—that is, the incidence of babies who were compromised but who had normal testing—was low and ranged from a PMR of 6.9 per 1000 for infants with normal AFV to 12.8 per 1000 for fetuses with a reactive NST. These investigators found that in most cases, the ultrasound-derived BPP parameters and NST could be completed within a relatively short time; each requires about 10 minutes.

Manning and colleagues¹⁰⁴ presented their experience with 26,780 high-risk pregnancies followed with the BPP. In their protocol, a routine NST was not performed if all of the ultrasound parameters were found to be normal for a score of 8. An NST was performed when one ultrasound finding was abnormal. The corrected PMR in this series was 1.9 per 1000 with fewer than 1 fetal death per 1000 patients within 1 week of a normal profile. Of all patients tested, almost 97% had a score of 8, which means that only 3% required further evaluation for scores of 6 or less. In a study of 525 patients with

scores of 6 or less, poor perinatal outcome was most often associated with either a nonreactive NST and absent fetal tone or a nonreactive NST and absent FBM.¹⁰⁵ A significant inverse linear relationship was observed between the last BPP score and both perinatal morbidity and mortality (Figs. 27.21 and 27.22).¹⁰⁶ Depending on the endpoint used, the false-positive rate ranged from 75% for a score of 6 to less than 20% for a score of 0. Manning has summarized the data reported in eight investigations using the BPP for fetal evaluation. Overall, 23,780 patients and 54,337 tests were reviewed. The corrected PMR, excluding lethal anomalies, was 0.77 per 1000.

The BPP correlates well with fetal acid-base status. Vintzileos and associates¹⁰¹ studied 124 patients undergoing cesarean birth before the onset of labor. Deliveries were undertaken for severe pre-eclampsia, elective repeat cesarean delivery, growth restriction, breech presentation, placenta previa, and fetal macrosomia. Acidosis was defined as an umbilical cord arterial pH of less than 7.20. The earliest manifestations of fetal acidosis were a nonreactive NST and loss of FBM. With scores of 8 or more, the mean arterial pH was 7.28, and

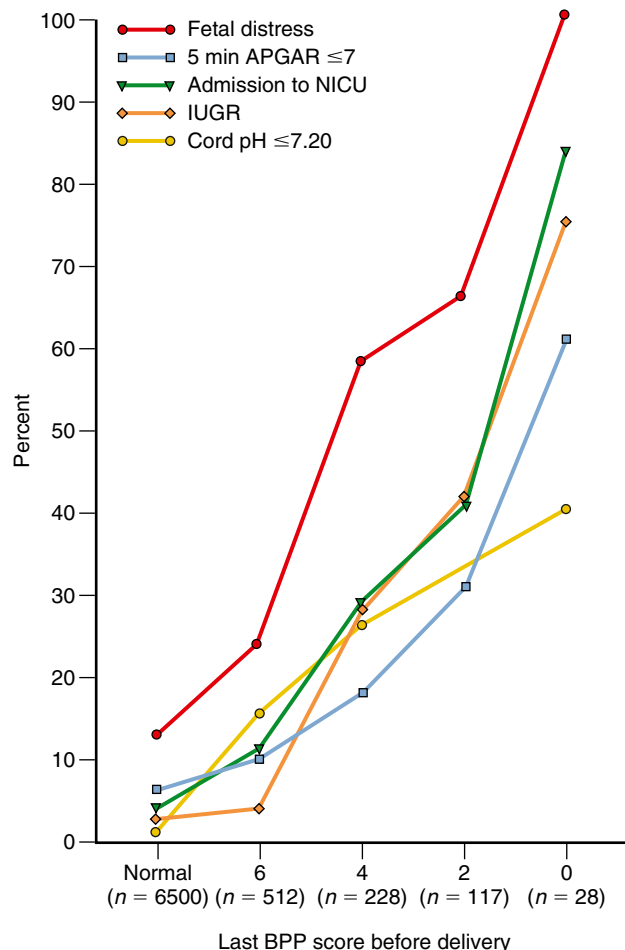


Fig. 27.21 Relationship Between Five Indexes of Perinatal Morbidity and Last Biophysical Profile (BPP) Score Before Delivery. A significant inverse linear correlation is observed for each variable. *IUGR*, Intrauterine growth restriction; *NICU*, neonatal intensive care unit. (From Manning FA, Harman CR, Morrison I, et al. Fetal assessment based on fetal biophysical profile scoring. *Am J Obstet Gynecol.* 1990;162:703.)

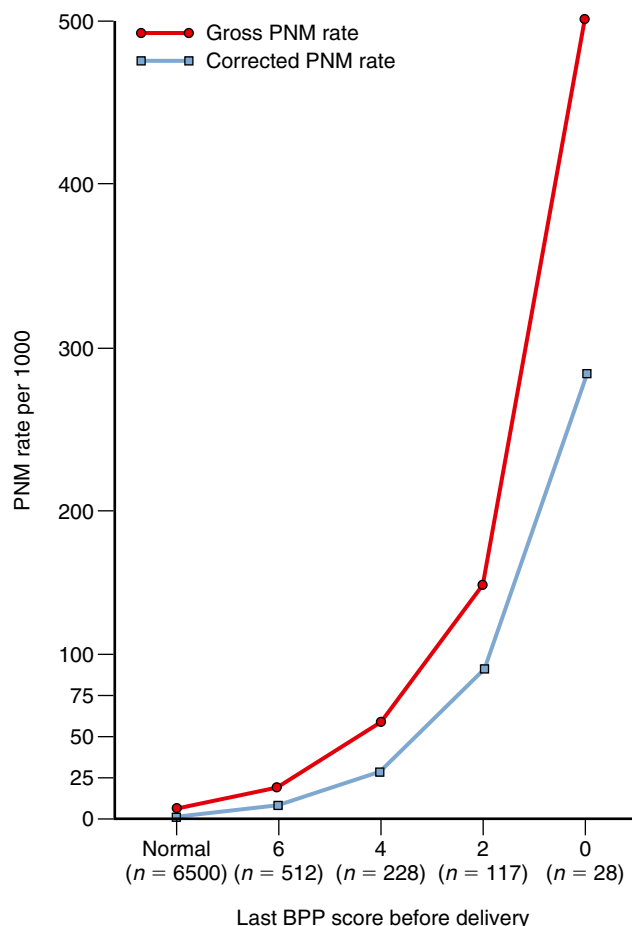


Fig. 27.22 Relationship Between Perinatal Mortality (PNM), Both Total and Corrected for Major Anomalies, and the Last Biophysical Profile (BPP) Score Before Delivery. A highly significant inverse and exponential relationship is observed. (From Manning FA, Harman CR, Morrison I, et al. Fetal assessment based on fetal biophysical profile scoring. *Am J Obstet Gynecol.* 1990;162:703.)

only 2 of 102 fetuses were acidotic. Nine fetuses with scores of 4 or less had a mean pH of 6.99, and all were acidotic.

Some studies have demonstrated that **antenatal corticosteroid administration may have an effect on the BPP, decreasing the profile score**. Because corticosteroids are used in cases of anticipated premature delivery (24 to 34 weeks), any false-positive results on biophysical testing may lead to inappropriate intervention and delivery. Kelly and co-workers reported that BPP scores were decreased in more than one-third of the fetuses tested at 28 to 34 weeks' gestation. This effect was seen within 48 hours of corticosteroid administration. Neonatal outcome was not affected. Repeat BPPs within 24 to 48 hours were normal in cases in which the BPP score had decreased by 4 points. The most commonly affected variables were FBM and the NST. Other investigators have reported transient suppression of FBM and heart rate reactivity after corticosteroid administration at less than 34 weeks' gestation, with return of these parameters to normal by 48 to 96 hours after corticosteroid treatment. This effect must be considered at institutions where daily BPPs are used to evaluate the fetus in cases of preterm labor or preterm premature rupture of the membranes (PPROM).

Some controversy exists regarding the utility of the BPP in predicting chorioamnionitis in pregnancies complicated by preterm

labor or PPROM. Sherer and colleagues reported that the absence of FBM is associated with histologic evidence of fetal inflammation and intrauterine infection in patients with preterm labor and intact membranes before 32 weeks' gestation. However, they recommend that this finding not be used to guide clinical management because of the low positive predictive value of absent FBM. Lewis and associates performed a randomized trial of daily NSTs vs. BPP in the management of PPROM. They concluded that neither daily NSTs nor BPPs had high sensitivity in predicting infectious complications in these patients. Daily BPPs increased cost without apparent benefit.

Manning and colleagues have described the correlation between biophysical scoring and the incidence of cerebral palsy in Manitoba. In patients referred for a BPP, an inverse, exponential, and highly significant relationship was found between last BPP score and the incidence of cerebral palsy. Scores of 6 or less had a sensitivity of 49%. The more abnormal the last BPP, the greater the risk for cerebral palsy. Gestational age, birthweight, and assumed timing of the injury were not related to the incidence of cerebral palsy, which ranged from 0.7 per 1000 live births for a normal BPP score to 13.1 per 1000 live births for a score of 6 to 333 per 1000 live births for a score of 0.

Several drawbacks of the BPP should be considered. Unlike the NST and CST, an ultrasound machine is required, and unless the BPP is video recorded, it cannot be reviewed. If the fetus is in a quiet sleep state, the BPP can require a long period of observation. The present scoring system does not consider the impact of polyhydramnios. For example, in a pregnancy complicated by diabetes mellitus, the presence of excessive amniotic fluid may be an indicator of fetal risk.

Predictive Value of the Biophysical Profile

To summarize the results of multiple studies, the false-negative rate of a normal BPP is less than 0.1%, or less than 1 fetal death per 1000 within 1 week of a normal BPP.² The false-positive rate of a particular test has always been of concern because of the possibility of unnecessary intervention (usually delivery) and subsequent iatrogenic complications. The BPP was developed in part to address the issue of the high false-positive rate of the CST and NST. Little attention has been paid to the possible false-positive rate of the abnormal or equivocal BPP. This is particularly relevant because the BPP is most commonly used as the final backup test in the NST and CST sequence of testing and is critically important when dealing with the premature fetus. As noted earlier, the false-positive rate of a score of 0 is less than 20%, but for a score of 6, it is up to 75%. Inglis and coworkers used VAS to define fetal condition with BPP scores of 6 or less in 81 patients at 28 to 42 weeks. Obstetric and neonatal outcomes of 41 patients whose score improved to normal after VAS were compared with those of 238 patients who had normal scores without VAS. The obstetric and neonatal outcomes were not significantly different between the two groups. VAS improved the BPP in about 80% of cases. Use of VAS for an equivocal BPP did not increase the false-negative rate and may reduce the likelihood of unnecessary obstetric intervention.

A 2008 Cochrane review¹⁰⁷ included evaluation of 2829 pregnancies randomized to BPP versus NST. The majority were term pregnancies. No differences were found in perinatal mortality, cesarean delivery rate, Apgar scores, or admission to the NICU. As with the 2015 Cochrane review⁸⁵ of the NST mentioned earlier, this analysis was underpowered to detect a significant difference in perinatal mortality among BPP and other modalities. The authors also note that “it is regrettable that since the introduction of the BPP in the 1980s, and following reports of observational studies of tens of thousands of pregnancies, fewer than 3000 women have been enrolled into randomized trials.”

Modified Biophysical Profile

In an attempt to simplify and reduce the time necessary to complete testing, a variety of modifications of the full BPP have been evaluated by focusing on the components of the BPP that are most predictive of perinatal outcome. The NST, an indicator of present fetal condition, may be combined with assessment of AFV, a marker of long-term status, in a modified BPP (mBPP). In this setting, a deepest vertical pocket (DVP) of amniotic fluid greater than 2×1 cm is usually considered normal, although different criteria have been applied. **Multiple investigations**, including a trial by Miller and colleagues of 56,617 antepartum tests in 15,482 pregnancies, **have demonstrated comparable results of the mBPP to the complete BPP, namely a false-negative rate (or rate of fetal death within 1 week of a normal mBPP) of 0.8 per 1000.** Nageotte and associates¹⁰⁸ demonstrated that the mBPP was as good a predictor of adverse fetal outcome as a negative CST. Additional evaluation for an abnormal mBPP is required in about 10% of patients. If the NST is nonreactive despite VAS or extended monitoring, or if the AFV is abnormal, either a full BPP or CST is performed. The CST as a backup test is associated with a higher rate of

intervention for an abnormal test than the use of a complete BPP as a backup test. Overall, the mBPP has a false-positive rate comparable with that of the NST but higher than that of the CST and full BPP. The low false-negative rate and ease of performance of the mBPP make it an excellent approach for the evaluation of large numbers of high-risk patients. As such, although potentially still a useful test, the CST has become less frequently used in current practice.

Whether to use a full AFI for AFV assessment or an abbreviated measure, the DVP, has undergone investigation. Chauhan and colleagues performed a randomized trial of more than 1000 women and found that the AFI led to more diagnoses of oligohydramnios, a higher rate of intervention, and more iatrogenic prematurity but offered no advantage in detecting or preventing adverse outcomes. Similarly, a 2009 Cochrane review¹⁰⁹ of five trials that included more than 3200 pregnancies concluded that no differences were apparent in perinatal outcomes. However, the use of AFI did result in an increased rate of diagnosis of oligohydramnios and an increased rate of induction of labor compared with DVP. This review could not comment on the relative ability of AFI, compared with DVP, for prevention of perinatal death because no deaths were reported in the included trials. On the basis of these data, endorsement of DVP as the primary fluid assessment strategy has been proposed.²¹

Doppler Ultrasound

The advent of Doppler ultrasound has permitted noninvasive assessment of the fetal, maternal, and placental circulations. With Doppler ultrasound, we can obtain information about uteroplacental blood flow and resistance, which may be markers of fetal adaptation and reserve. **This method of fetal assessment has only been demonstrated to be of value in reducing perinatal mortality and unnecessary obstetric interventions in fetuses with suspected FGR and possibly other disorders of uteroplacental blood flow.²** A detailed description of the underlying principles and use of Doppler ultrasound for fetal assessment of FGR is available in [Chapter 30](#). For the purposes of this chapter on antenatal fetal assessment, Doppler interrogation of fetal vascular flow and resistance can be conceptualized as a follow-up test to determine fetal reserve in cases of suspected FGR and not as a primary method of antenatal fetal surveillance for either high- or low-risk pregnancies.

Perhaps ironically, however, Doppler ultrasound has been more stringently evaluated in randomized trials than other antenatal testing methods, as has been summarized in an editorial by Divon and Ferber. The summary of available evidence from a 2017 Cochrane review of 19 randomized trials included more than 10,000 high-risk pregnancies, in which the use of Doppler interrogation of the umbilical artery was associated with decreased perinatal deaths (RR, 0.71; 95% CI, 0.52 to 0.98) and significantly fewer inductions of labor and cesarean deliveries. Doppler interrogation of the ductus venosus showed a similar trend but was not conclusive for benefit in routine surveillance of high-risk pregnancies. Studies of low-risk pregnancies have not shown a benefit from the use of Doppler ultrasound, as described in a 2015 systematic review of five studies that included more than 14,000 pregnancies.

CLINICAL APPLICATION OF TESTS OF FETAL WELL-BEING

Our ability to detect and prevent impending fetal death or injury depends on not only the predictive value of the tests used and the population selected for testing but also on our ability to respond to abnormal test results. To have an impact on testing outcomes and the overall fetal death rate, we must consider available strategies to deal

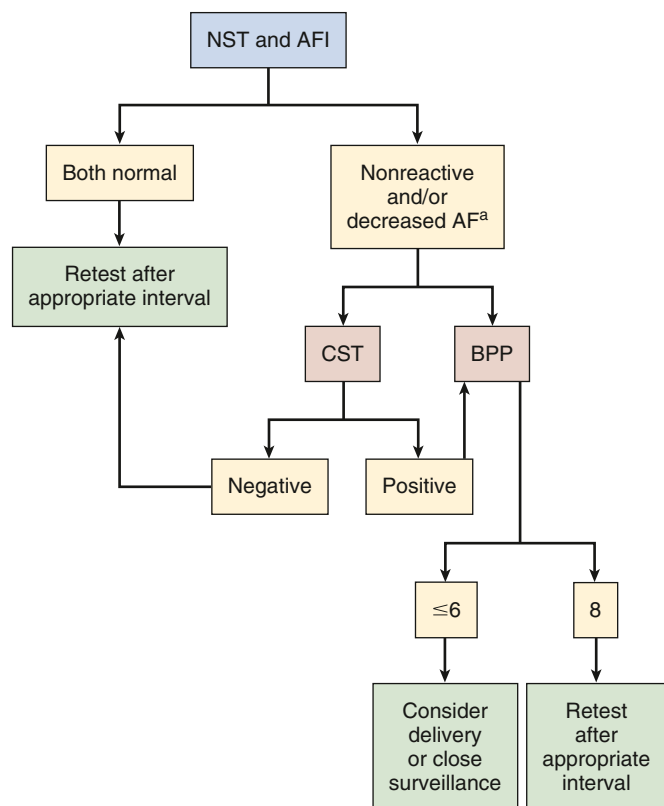


Fig. 27.23 Flow Chart for Antepartum Fetal Surveillance. The non-stress test (NST) and amniotic fluid index (AFI) are used as the primary methods for fetal evaluation. A nonreactive NST and decreased AFI are further evaluated using either the contraction stress test (CST) or the biophysical profile (BPP). Further details regarding the use of the BPP are provided in Table 27.6.

^aIf the fetus is mature and amniotic fluid volume is reduced, delivery should be considered before further testing is undertaken. (Modified from Finberg HJ, Kurtz AB, Johnson RL, et al. The biophysical profile: a literature review and reassessment of its usefulness in the evaluation of fetal well-being. *J Ultrasound Med.* 1990;9:583.)

with abnormal test results. These strategies would ideally include a series of antenatal evaluations and interventions short of premature delivery, with premature delivery ultimately reserved for cases when it is ascertained that intrauterine injury or death can no longer be delayed or prevented. Fig. 27.23 presents a practical testing strategy, which includes the use of combinations of antepartum tests in an organized sequence to further evaluate the fetus who displays an initial abnormal test. For suspected fetal compromise, follow-up steps may include hospitalization for more intensive monitoring, administration of antenatal steroids, and identification/correction of maternal or fetal factors potentially contributing to interruptions in the fetal oxygenation pathway.

Testing can be initiated at early gestational ages in high-risk pregnancies (25 to 26 weeks) to identify the fetus at risk. Maternal and fetal interventions can then be considered, with appropriate caution about the high false-positive rate of testing and costly potential effects of recommending a preterm birth at that gestational age. Obviously, safe prolongation of intrauterine life is the primary goal, and better understanding of the pathophysiology of the premature fetus and the use of combinations of tests will allow this to be accomplished.

The question of routine antepartum fetal surveillance must be carefully examined. Antepartum fetal testing can more accurately predict fetal outcome than antenatal risk assessment using an established scoring system. **Patients judged to be at high risk based on known medical factors but whose fetuses demonstrate normal antepartum fetal evaluation have a lower PMR than patients considered at low risk whose fetuses have abnormal antepartum testing results.** Routine antepartum fetal evaluation would be necessary to detect the considerable proportion of fetuses destined for fetal death or injury with no identifiable risk factors. It would therefore seem reasonable to consider extending some form of antepartum fetal surveillance to all obstetric patients, such as maternal assessment of fetal activity as described previously.

How best to combine antenatal tests in clinical practice and to determine which patients should be eligible remains a challenge. **The approach to prescribing testing modalities must take into account gestational age, medical comorbidities, and sociodemographic risk factors** described in this chapter in order to prevent fetal injury or death without causing iatrogenic prematurity or an excess of testing and worry.

The condition-specific use of antenatal testing may be illustrated by the following examples. In a prolonged pregnancy, a **parallel testing scheme** would be used. In this situation, the obstetrician is not concerned with fetal maturity but rather fetal well-being. Several tests are performed at the same time, such as antepartum fetal heart rate testing and the BPP. It is acceptable in this high-risk situation to intervene when a single test is abnormal. It seems prudent to accept a false-positive test result to avoid the intrauterine death of a mature and otherwise healthy fetus. In most other high-risk pregnancies, such as those complicated by diabetes mellitus or hypertension, it is preferable to allow the fetus to remain in utero as long as possible. In these situations, a **branched testing scheme** is used to decrease the likelihood of unnecessary premature intervention. The obstetrician uses a series of tests and, under most circumstances, would only deliver a premature infant when all parameters suggest fetal compromise. In this situation, the physician must consider the likelihood of neonatal respiratory distress syndrome (RDS) and other short- and long-term consequences of prematurity and review these risks with the patient in consultation with neonatology experts.

The comparable performance of the various available antenatal tests, along with their utility in series, has been illustrated by numerous investigators (see Table 27.2). Maternal assessment of fetal activity would appear to be a reasonable first-line screening test for both high- and low-risk patients. The use of this approach may decrease the number of unexpected intrauterine deaths in so-called normal pregnancies. The NST and mBPP remain the primary methods used for antepartum fetal evaluation in high-risk patients at most centers, with full BPP and CST to assess fetal condition in patients who exhibit a persistently nonreactive NST or abnormal mBPP. This sequential approach may be particularly valuable in avoiding unnecessary premature intervention. The NST and mBPP can be quickly performed in an outpatient setting and are easily interpreted. In contrast, the CST is usually performed on or near the labor and delivery suite, it may require an intravenous infusion of oxytocin, and it may be more difficult to interpret. However, the ability of the CST to stress the fetus and evaluate its response to intermittent interruptions in intervillous blood flow may provide an earlier warning of fetal compromise than the NST or mBPP.

The frequency with which to use specific tests will depend on a number of features, including the predictive value of the test and the underlying condition prompting the test. Most tests described previously have relatively reliable risk profiles over the course of a

week when normal, with decreased intervals between tests recommended for abnormal results. Regarding the underlying condition contributing to risk for adverse outcomes, consideration must be given to whether that condition is stable, worsening, or improving. The gestational age at which to initiate testing has not been clearly defined for most high-risk conditions. **Initiating testing at 32 to 34 weeks' gestation has historically been prescribed for most high-risk pregnancies, with earlier testing recommended for cases with multiple comorbidities or particularly worrisome features.**

In considering the pros and cons of testing in the preterm period, it is worthwhile to return to the question of potential harms of testing, from increases in preterm birth and cesarean delivery rates to maternal anxiety and financial costs. Although no evidence exists of improved outcomes from any one testing strategy, as a counterpoint some authors have begun to study whether the implementation of testing increases the chances of induction of labor or cesarean delivery rates. In addition, it is reasonable to question whether a program of antenatal testing contributes to maternal anxiety or other indirect costs that are difficult to quantify. Taking patient-specific risk factors, and perhaps maternal preferences, into account when considering an individualized testing plan is most likely to contribute to optimal outcomes.

On behalf of the Eunice Kennedy Shriver National Institute of Child Health and Human Development Workshop, Signore and co-workers² proposed a series of guidelines for condition-specific antenatal testing based on the available evidence. In their 2009 publication, they stress that **"the basis for antepartum testing relies on the premise that the fetus whose oxygenation in utero is challenged will respond with a series of detectable physiologic adaptive or decompensatory signs as hypoxemia or frank metabolic acidemia develop,"** hence the recommendation for the use of antenatal tests in series to follow the changes in observable measures of fetal response to a suboptimal intrauterine environment. Similarly, updated assimilation of condition-specific risks into a proposed testing scheme was conducted by ACOG and the Society for Maternal Fetal Medicine in deriving a set of testing recommendations in 2021 that expands on the NICHD scheme (Table 27.6).²¹ Although this proposed strategy is certainly not comprehensive, it is hoped that future research into specific testing strategies stratified by risk category will enhance our ability to prescribe these tests effectively and safely.

Evidence for Condition-Specific Testing

For most of the conditions identified as constituting an increased risk for fetal death, insufficient studies exist to permit an evidence-based recommendation for a particular testing scheme. In addition to this limitation, condition-specific testing is problematic as a general strategy for the prevention of fetal death, given the many fetal deaths that occur in pregnancies otherwise categorized as low risk or without identifiable risk factors.² However, it is incumbent on care providers to consider all conditions associated with an increased risk for fetal death or other adverse outcomes as a potential indication for some form of antenatal surveillance^{2,21} and to individualize antenatal testing plans based on specific underlying conditions. This strategy is outlined in an excellent summary by Kontopoulos and Vintzileos,¹¹⁰ in which they explore several pathophysiologic mechanisms as potentially distinct etiologies for risk of fetal death in different populations. **The authors acknowledge that no ideal single test or testing strategy exists for all high-risk pregnancies but that clinician judgment and logic, as well as evidence from observational studies, should guide testing strategies for each patient.**

Difficulty in generating evidence-based recommendations for condition-specific testing schemes in cases of identified risk factors can be illustrated by considering the example of hypertensive disorders. Well-designed prospective trials that provide evidence for any specific testing strategy are not available, despite the frequent use of a variety of antenatal testing schemes used in clinical practice for hypertensive disorders. The heterogeneity of pathophysiologic mechanisms at play in the general category of hypertensive disorders contributes to the difficulty in generating specific recommendations. Although varying iterations of expert recommendations do not support antepartum fetal testing for mild to moderate blood pressure abnormalities, nor do they support it in the absence of preeclampsia or FGR, for many practitioners the evidence that all forms of chronic hypertension constitute an increased risk for fetal death and adverse outcomes prompts broad use of testing modalities for these patients.

Another challenge in the implementation of condition-specific testing may lie in the appreciation and acknowledgment of individual patient characteristics as risk factors for fetal death, in particular certain demographic risk factors without clear pathophysiologic links. For example, the condition of advanced maternal age has garnered a great deal of investigation into associations with fetal death and other adverse pregnancy outcomes and related management strategies. However, the same cannot be said for other demographic risk factors, such as obesity and Black race (via exposure to racism), which have equal or greater relative risks of fetal death yet historically have been less accepted in the community as risks for adverse outcomes warranting examination of disparities and increased antenatal surveillance.

ASSESSMENT OF FETAL PULMONARY MATURATION

This section reviews those techniques that enable the obstetrician to predict the risks of RDS for the infant who requires premature delivery. The risk of iatrogenic prematurity for the neonate balanced against the risk of continued antepartum assessment of the potentially compromised fetus determines ultimate management strategies. As one marker of fetal maturity that can be assessed antenatally, incorporation of the discussion of fetal pulmonary maturity assessment with that of antepartum assessment has historical relevance. However, we have evolved an appreciation that the multifactorial nature of preterm neonatal adaption is hard to predict on the basis of pulmonary maturity testing alone. Among other factors, this paradigm has decreased the frequency with which this testing is employed in clinical practice in association with abnormal antepartum fetal tests, when those tests indicate preterm delivery to prevent fetal death. Important differences in neonatal adaptation, including development of RDS, have been demonstrated between preterm and term neonates and also with each advancing gestational week from 37 to 39 weeks. This places even more responsibility on the physician to judiciously assess indications for and timing of delivery before 39 weeks. Nevertheless, in weighing the risks and benefits of effecting delivery prompted by antenatal testing results before 39 weeks, assessment of fetal pulmonary maturity is rarely indicated.

Tests of Fetal Pulmonary Maturity

Available methods for evaluating fetal pulmonary maturity rely generally on either the presence or quantitation of components of

TABLE 27.6 Proposed Testing Strategy by Selected Factors Associated With Increased Risk of Stillbirth

Factor	Suggested Gestational Age to Begin Antenatal Fetal Surveillance	Suggested Frequency of Antenatal Fetal Surveillance
Fetal		
Growth restriction ^a		
UAD: normal or with elevated impedance to flow in umbilical artery with diastolic flow present; with normal AFI and no other concurrent maternal or fetal conditions	At diagnosis ²	Once or twice weekly
UAD: AEDV or concurrent conditions (oligohydramnios, maternal comorbidity (e.g., preeclampsia, chronic hypertension)	At diagnosis ²	Twice weekly ³ or consider inpatient management
UAD: REDV	At diagnosis ^b	Inpatient management ^c
Multiple gestation		
Twins, uncomplicated dichorionic	36 ^{0/7} weeks	Weekly
Twins, dichorionic, complicated by maternal or fetal disorders, such as fetal growth restriction	At diagnosis ^b	Individualized
Twins, uncomplicated monochorionic-diamniotic	32 ^{0/7} weeks ^d	Weekly
Twins, complicated monochorionic-diamniotic (i.e., TTTS)	Individualized	Individualized
Twins, monoamniotic	Individualized	Individualized
Triplets and higher-order multiples	Individualized	Individualized
Decreased fetal movement	At diagnosis ^c	Once ^e
Fetal anomalies and aneuploidy	Individualized	Individualized
Maternal		
Hypertension, chronic		
Controlled with medications	32 ^{0/7} weeks	Weekly
Poorly controlled or with associated medical conditions	At diagnosis ^b	Individualized
Gestational hypertension/preeclampsia		
Without severe features	At diagnosis ^{b,c}	Twice weekly
With severe features	At diagnosis ^{b,c}	Daily
Diabetes		
Gestational, controlled on medications without other comorbidities	32 ^{0/7} weeks	Once or twice weekly
Gestational, poorly controlled	32 ^{0/7} weeks	Twice weekly
Pregestational	32 ^{0/7} weeks ^f	Twice weekly
Systemic lupus erythematosus		
Uncomplicated	By 32 ^{0/7} weeks	Weekly
Complicated ^g	At diagnosis ^b	Individualized
Antiphospholipid syndrome	By 32 ^{0/7} weeks ^h	Twice weekly
Sickle cell disease		
Uncomplicated	32 ^{0/7} weeks	Once or twice weekly
Complicated ⁱ	At diagnosis ^b	Individualized
Hemoglobinopathies other than Hb SS disease	Individualized	Individualized
Renal disease (Cr greater than 1.4 mg/dL)	32 ^{0/7} weeks	Once or twice weekly
Thyroid disorders, poorly controlled	Individualized	Individualized
In vitro fertilization	36 ^{0/7} weeks	Weekly
Substance use		
Alcohol, 5 or more drinks per week	36 ^{0/7} weeks	Weekly
Polysubstance use	Individualized	Individualized
Prepregnancy BMI		
Prepregnancy BMI 35.0–39.9 kg/m ²	37 ^{0/7} weeks	Weekly
Prepregnancy BMI 40 kg/m ² or above	34 ^{0/7} weeks	Weekly
Maternal age older than 35 years	Individualized ^j	Individualized

TABLE 27.6 Proposed Testing Strategy by Selected Factors Associated With Increased Risk of Stillbirth—cont'd

Factor	Suggested Gestational Age to Begin Antenatal Fetal Surveillance	Suggested Frequency of Antenatal Fetal Surveillance
Obstetric		
Previous stillbirth		
At or after 32 ^{0/7} weeks	32 ^{0/7} weeks ^k	Once or twice weekly
Before 32 ^{0/7} weeks of gestation	Individualized	Individualized
History of other adverse pregnancy outcomes in immediately preceding pregnancy		
Previous fetal growth restriction requiring preterm delivery	32 ^{0/7} weeks	Weekly
Previous preeclampsia requiring preterm delivery	32 ^{0/7} weeks	Weekly
Cholestasis	At diagnosis ^b	Once or twice weekly
Late term	41 ^{0/7} weeks	Once or twice weekly
Abnormal serum markers ^l		
PAPP-A less than or equal to the fifth percentile (0.4 MoM)	36 ^{0/7} weeks	Weekly
Second-trimester Inhibin A equal to or greater than 2.0 MoM	36 ^{0/7} weeks	Weekly
Placental		
Chronic placental abruption ^m	At diagnosis ^b	Once or twice weekly
Vasa previa	Individualized	Individualized
Velamentous cord insertion	36 ^{0/7} weeks	Weekly
Single umbilical artery	36 ^{0/7} weeks	Weekly
Isolated oligohydramnios (single deepest vertical pocket <2 cm)	At diagnosis ^{b,c}	Once or twice weekly
Polyhydramnios, moderate to severe (deepest vertical pocket ≥12 cm or AFI ≥30 cm)	32 ^{0/7} –34 ^{0/7} weeks ⁿ	Once or twice weekly

^aEstimated fetal weight or abdominal circumference less than the 10th percentile.

^bOr at a gestational age when delivery would be considered because of abnormal test results.

^cIf not delivered.

^dIn addition to routine surveillance for twin-to-twin transfusion syndrome and other monochorionic twin complications.

^eRepeat if decreased fetal movement recurs.

^fOr earlier for poor glycemic control or end-organ damage.

^gSuch as active lupus nephritis, recent lupus flare, antiphospholipid antibodies with prior fetal loss, anti-RO/SSA or anti-La/SSB antibodies, or thrombosis.

^hIndividualize; take into consideration obstetric history, number of positive antibodies, and current pregnancy complications.

ⁱSuch as maternal hypertension, vasooclusive crisis, placental insufficiency, fetal growth restriction.

^jBased on cumulative risk when present with other factors.

^kOr starting 1–2 weeks before the gestational age of the previous stillbirth.

^lIf serum screening for aneuploidy is performed, the results may be considered in determining whether antenatal fetal surveillance should be performed.

^mIn individuals who are candidates for outpatient management.

ⁿOr at diagnosis if diagnosed after 32^{0/7}–34^{0/7} weeks.

AEDV, Absent end diastolic velocity; AFI, amniotic fluid index; BMI, body mass index; Cr, creatinine; MoM, multiples of the median; PAPP-A, pregnancy-associated plasma protein A; REDV, reversed end diastolic flow; S/D, systolic-to-diastolic; TTTS, twin-to-twin transfusion syndrome; UAD, umbilical artery Doppler.

Modified from Indications for outpatient antenatal fetal surveillance. ACOG Committee Opinion No. 828. American College of Obstetricians and Gynecologists. *Obstet Gynecol.* 2021;137:e177-97.

pulmonary surfactant or measurement of surfactant function. The former category has become the most common and reliable in current practice, although no data show that one method is preferable to the others with regard to prediction of pulmonary maturity. **In general, a test that is positive for fetal pulmonary maturity will much more accurately predict the absence of RDS than a negative test will predict the presence of RDS.**

With the exception of amniotic fluid specimens obtained from the vaginal pool, the evaluation of fetal pulmonary maturation requires that a sample of amniotic fluid be obtained by amniocentesis. This is generally a low-risk procedure with few potential adverse outcomes,

including unsuccessful amniocentesis (1.6% to 4.4%) and complications that require same-day delivery (0.7% to 3.3%).

Methods of quantifying pulmonary surfactant rely on assessing properties or ratios of the phospholipid content in amniotic fluid that typically emerge late in gestation, between 35 and 37 weeks. These include the lecithin/sphingomyelin (L/S) ratio, the presence of PG, visual inspection of fluid, and surfactant/albumin ratio. Some test results are influenced by the presence of blood or meconium in the amniotic fluid sample, such as the L/S ratio, whereas others, including PG assays, are not affected by these substances. Measurement of surfactant function can be estimated by performing a lamellar body

count. This test is relatively quick and easy to perform compared with the quantification methods noted earlier, and results are not affected by presence of blood nor meconium.

Determination of Fetal Pulmonary Maturation in Clinical Practice

A paradigm shift that has taken place over the years has seen a decrease in the frequency with which fetal lung maturity testing is recommended and used, as stated at the beginning of this section. Both ACOG and the NICHD^{21,57} have endorsed increased stringency of indications for late preterm and early term births, which, if followed, stand to decrease the opportunities for clinical uncertainty around the need for delivery and thus decrease the need for fetal lung maturity testing. As noted in the 2011 NICHD publication on this topic, "If significant maternal or fetal risk exists, delivery should occur regardless of biochemical maturity, and if delivery could be deferred owing to the absence of pulmonary maturity, there is not a stringent indication for prompt delivery. Additionally, it is recognized that a mature fetal lung profile denoting the presence of pulmonary surfactant does not necessarily translate to maturity of other organ systems."¹⁰⁶

The advent of increasingly routine administration of antenatal corticosteroids for the promotion of fetal lung maturity in the late preterm period between 34 and 37 weeks, based on a large multicenter randomized controlled trial of this intervention for the prevention of short-term adverse neonatal outcomes associated with delivery in this gestational period, will further decrease the frequency of fetal pulmonary maturity testing in routine clinical practice.

KEY POINTS

- Although the PMR has fallen steadily in the United States since 1965, the number of fetal deaths has not changed substantially in the past decade.
- Perinatal events play an important role in infant mortality and long-term disability of survivors in addition to their contribution to fetal death.
- At least 20% of fetal deaths have no obvious fetal, placental, maternal, or obstetric etiology, and this percentage increases with advancing gestational age.
- The prevalence of an abnormal condition (i.e., fetal death) has great impact on the predictive value of antepartum fetal tests.
- Few of the antepartum tests commonly used in clinical practice today have been subjected to large-scale prospective and randomized evaluations that can speak to the true efficacy of testing.
- Fetal adaptation to hypoxemia is mediated by changes in heart rate and redistribution of cardiac output.
- The decrease in fetal movement with hypoxemia makes maternal assessment of fetal activity a potentially simple and widely applicable method of monitoring fetal well-being. However, prospective trials of this method for the prevention of perinatal mortality have failed to conclusively show benefit.
- The CST has a low false-negative rate but a high false-positive rate and is cumbersome to perform; thus it is used less frequently in common practice than other testing modalities.
- The observation that accelerations of the fetal heart rate in response to fetal activity, uterine contractions, or stimulation reflect fetal well-being is the basis for the NST.
- Use of VAS for a nonreactive NST or equivocal BPP does not increase the false-negative rate and may reduce the likelihood of unnecessary obstetric intervention.
- The NST has a low false-negative rate, although higher than that of CST, and a high false-positive rate.
- Fetal biophysical activities can be evaluated with real-time ultrasonography by BPP, and those fetal biophysical activities that are present earliest in fetal development are the last to disappear with fetal hypoxia.
- The mBPP performs comparably with the full BPP. Both modalities have a false-negative rate, or rate of fetal death within 1 week of a normal test, of 0.8 per 1000.
- When delivery before 39 weeks is prompted by abnormal antepartum fetal test results, anticipated total neonatal maturity should influence decision making. Results of amniotic fluid tests of fetal maturation should be interpreted in this context.
- Condition-specific testing involves modifying the frequency, type, and initiation of antenatal tests according to maternal high-risk conditions.
- Both costs and benefits should be considered when using a particular testing strategy, taking into account competing risks of fetal death and postnatal morbidity.

SUMMARY

We must be aware of our limited understanding of the pathophysiology of fetal death and injury in many situations, and we must attempt judicious use of antenatal testing measures with the hope of preventing some adverse outcomes and with the goal of doing no harm. Although we assume that by testing and intervening with delivery for abnormal test results we are contributing to a good outcome, as Scifres and Macones⁶⁹ stated in their 2008 review of costs and benefits of antenatal testing, "In the absence of data, we cannot be assured that our interventions, even if they prevent stillbirth, may not involve a tradeoff between long-term neurologic dysfunction and fetal death." Additionally, aside from potential unknown medical sequelae of antenatal testing and interventions, monetary, time, and psychological costs must also be considered, but these are difficult to quantify in both research settings and in practice.

As clinicians aim to implement evidence-based strategies to screen for and prevent fetal injury and death, the limitations of existing research must be considered. Because it is in many ways impractical and perhaps unethical to carry out placebo-controlled trials of antenatal testing methodologies in high-risk pregnancies, we must acknowledge that this type of rigorous evidence is unlikely to be imminently forthcoming.² Nevertheless, we propose that biological plausibility and clinician judgment prevail over therapeutic nihilism in clinical practice and that future research efforts embrace a creative approach to both condition-specific and apparently unpredictable fetal death and injury.

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Visit <http://eBooks.health.elsevier.com/> for a complete list of references.

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